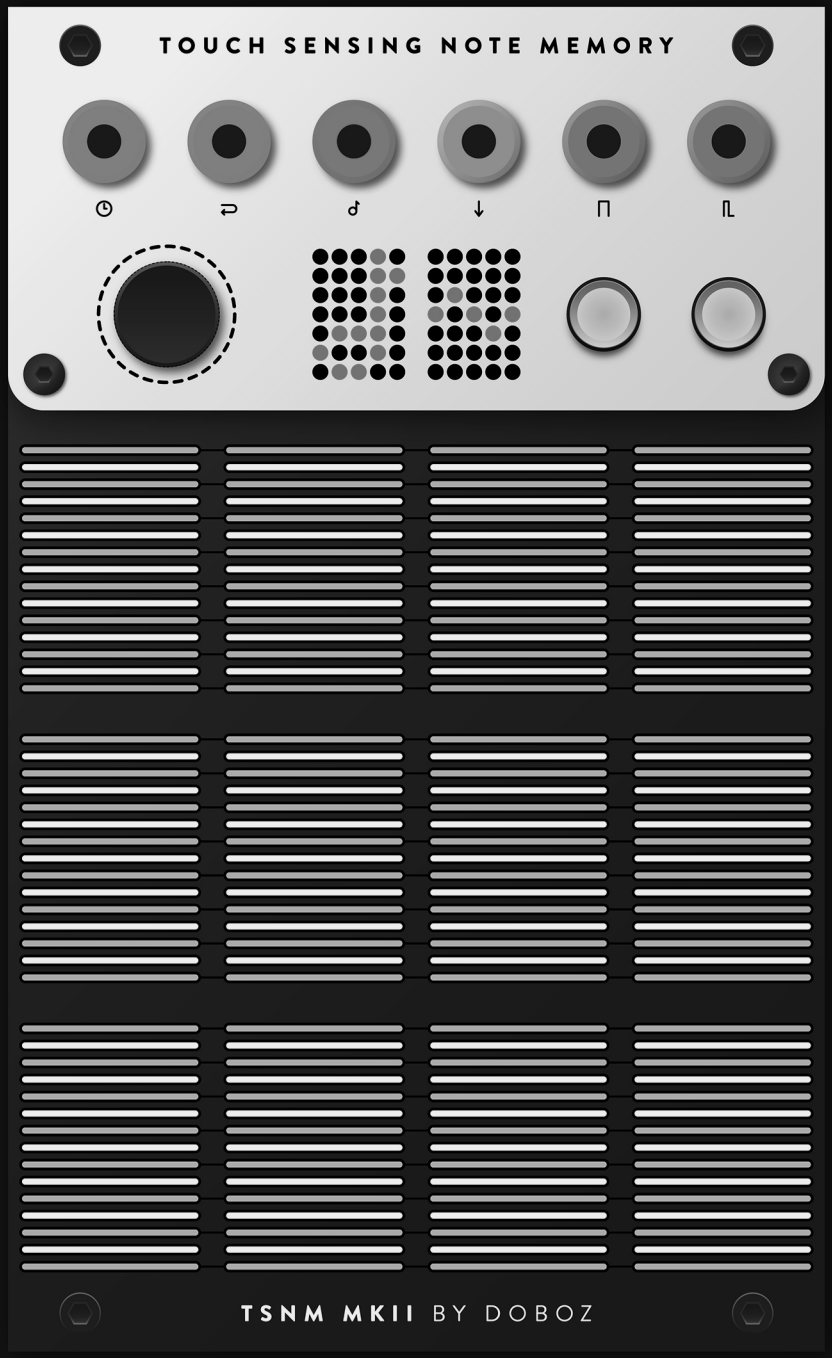

ASSEMBLY GUIDE

TSNM MKII 4U

Welcome to the assembly guide for the Touch Sensing Note Memory (MKII). This project is suitable for more advanced users. It is almost necessary to have basic soldering skills and to be able to identify electronic components. Before starting it look through this guide so that you'll be familiar with the building process and you won't run into any surprises!

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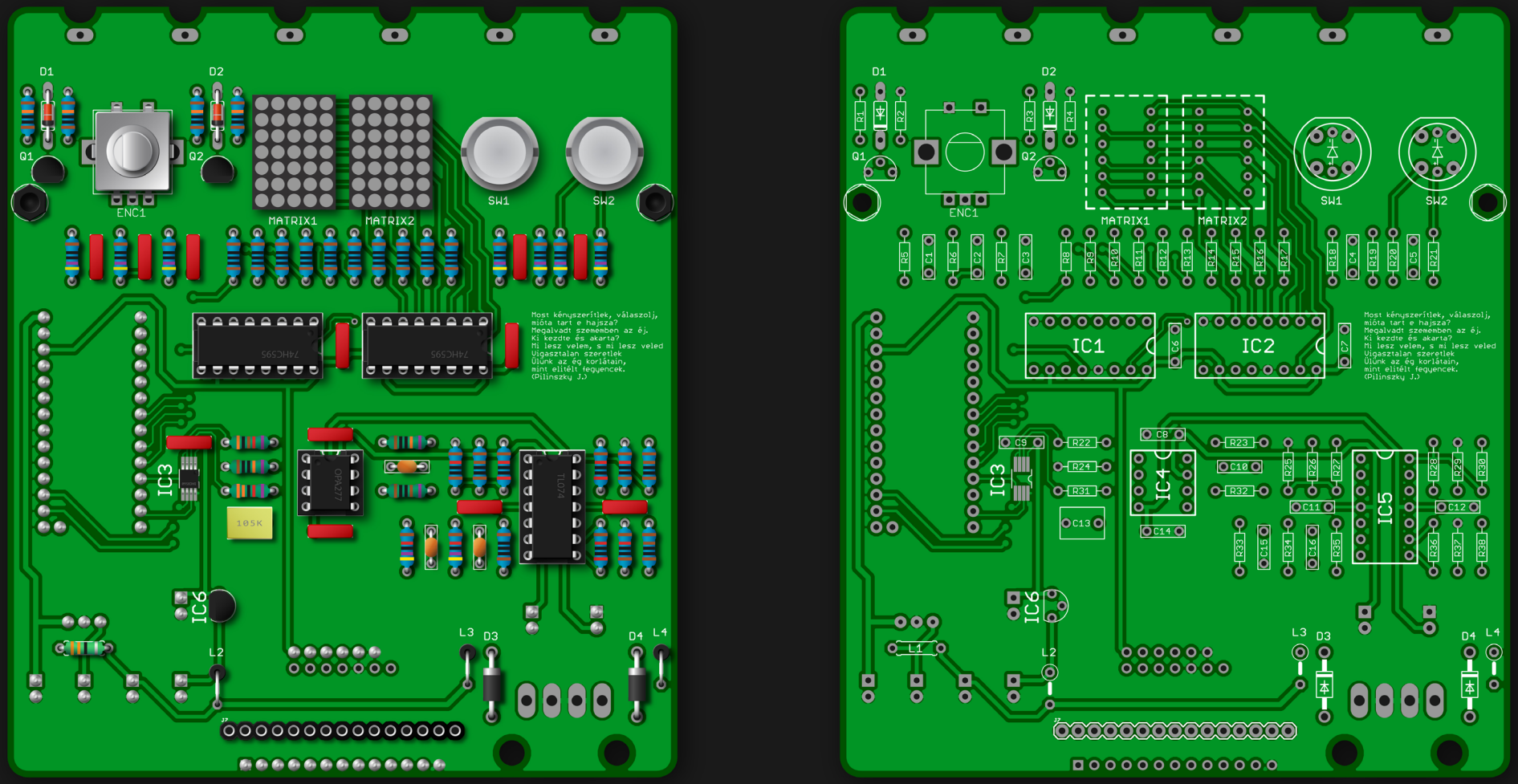
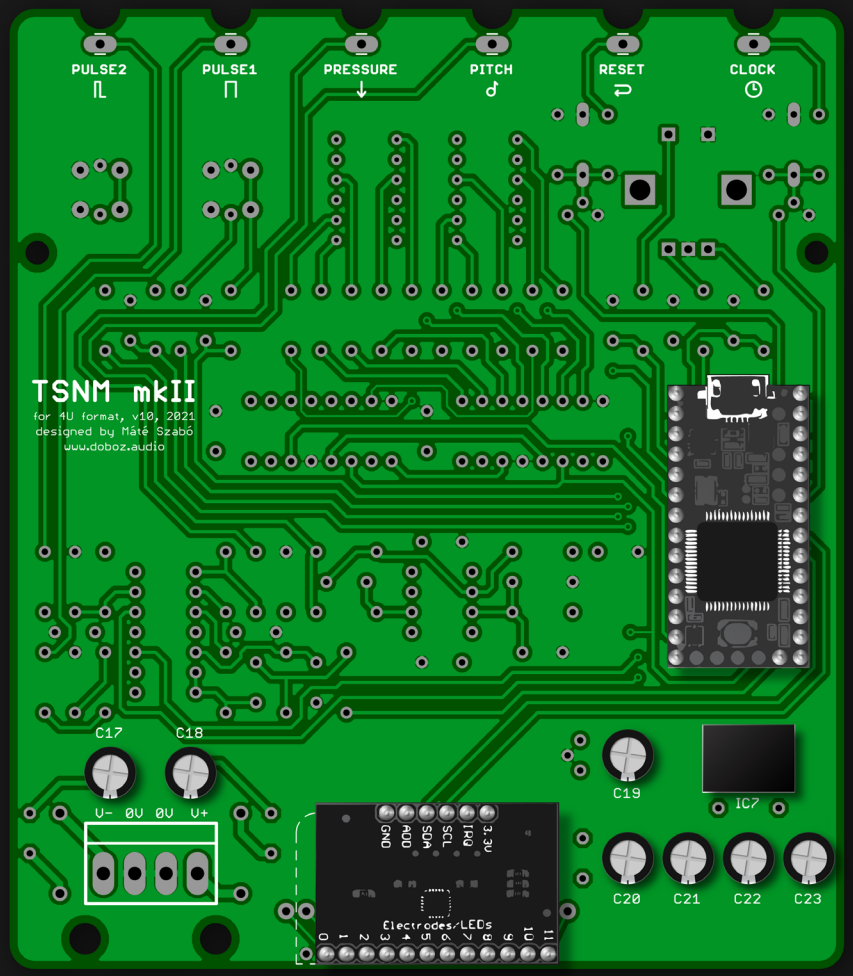
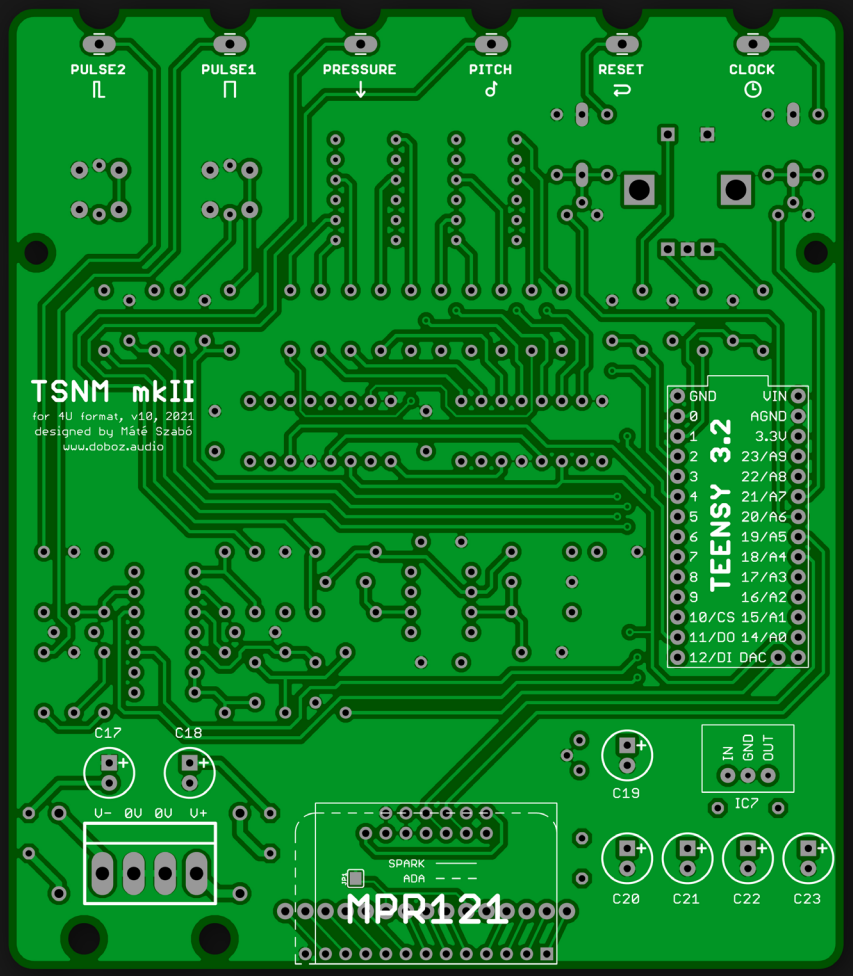
#22 VOLTAGE REGULATOR
#23 FERRITE BEADS
#24 ELECTROLYTIC CAPACITORS
#26 MISC. COMPONENTS
#37 FIRMWARE UPLOAD



TOP SIDE OF PCB





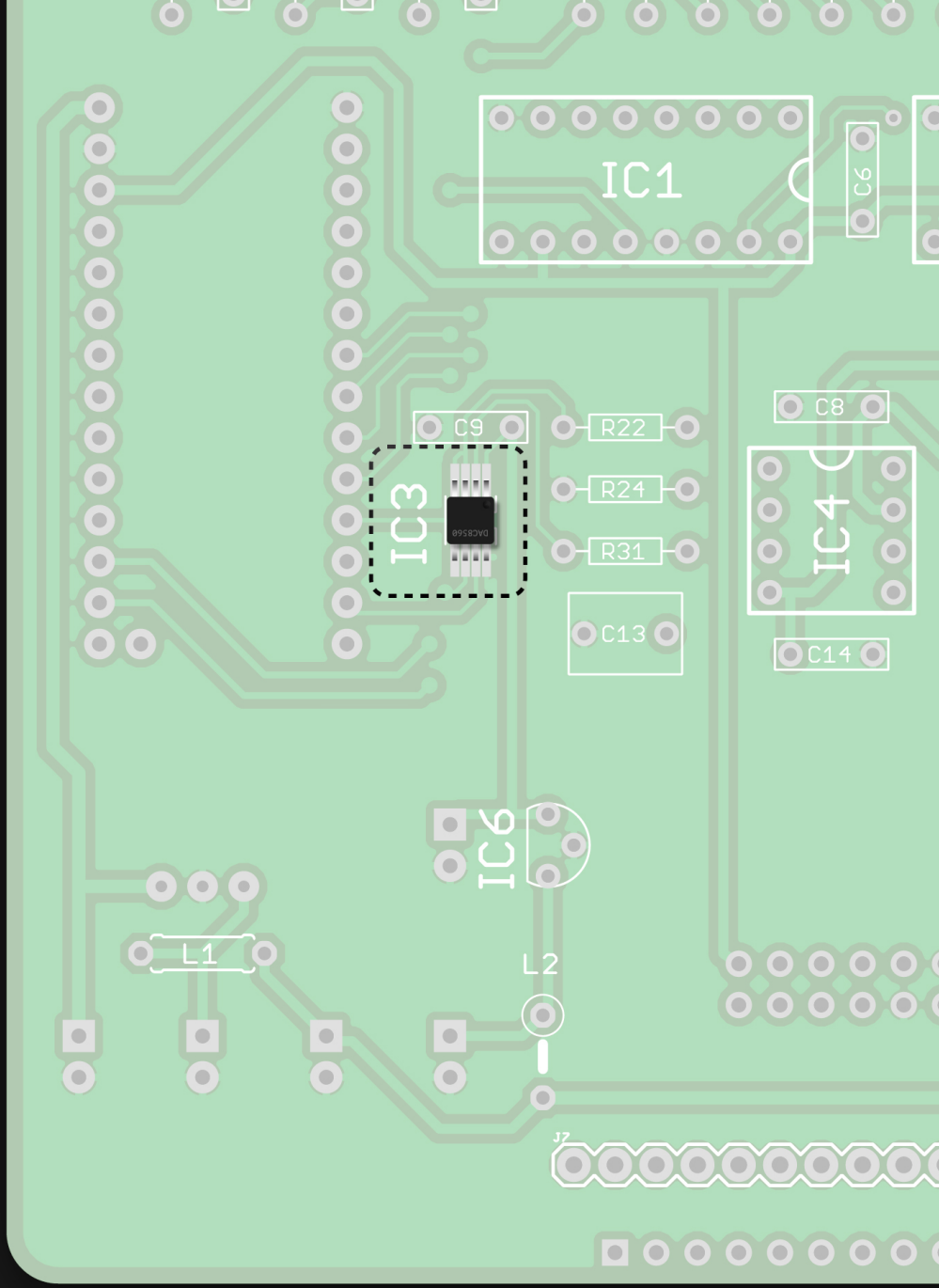
BOTTOM SIDE OF PCB

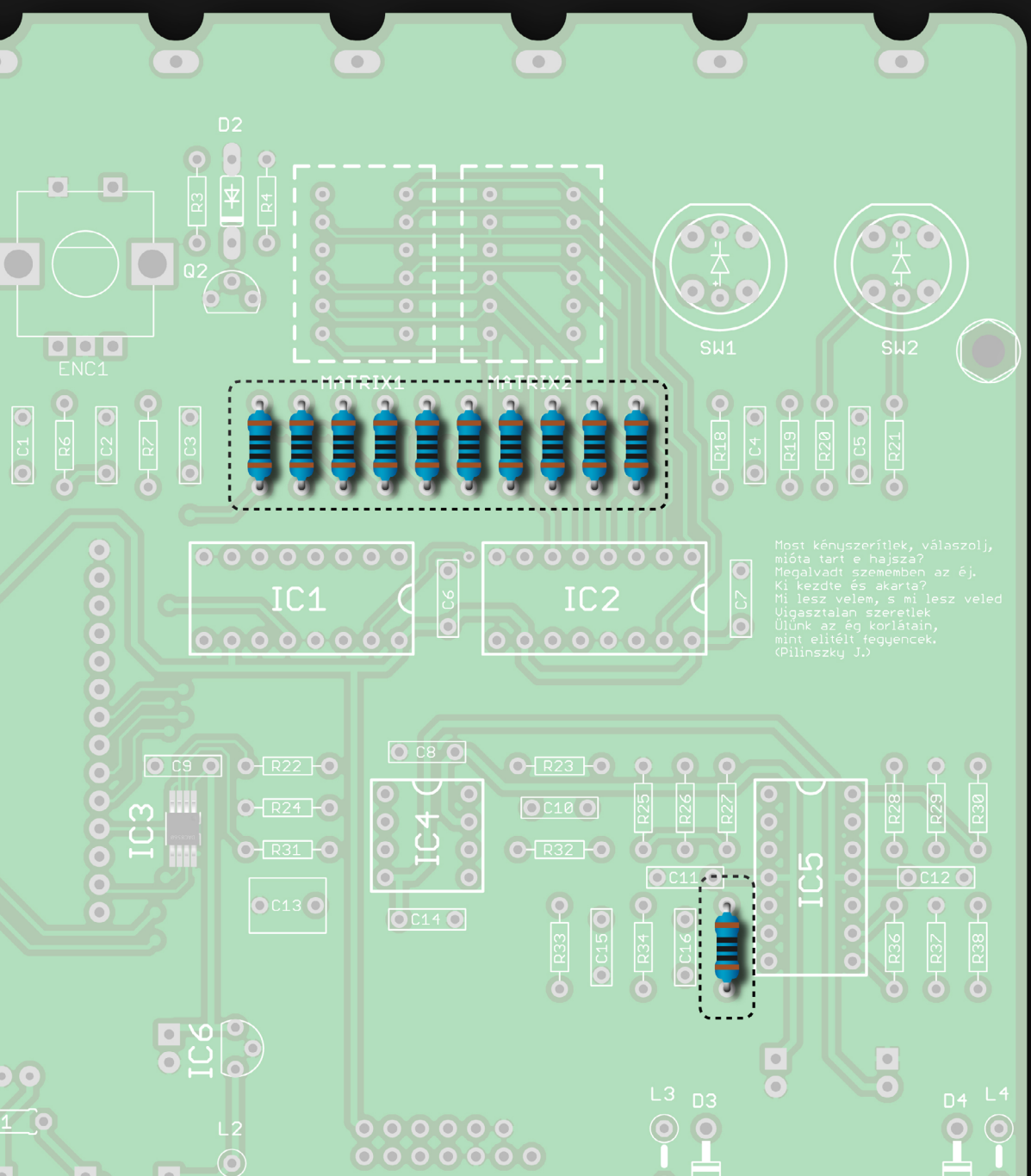
#01 / DIGITAL-TO-ANALOG CONVERTER

1x DAC8560

IC3

ORIENTATION: **THE CIRCULAR RECESS SHOULD BE IN THE TOP RIGHT CORNER (AS PICTURED)**



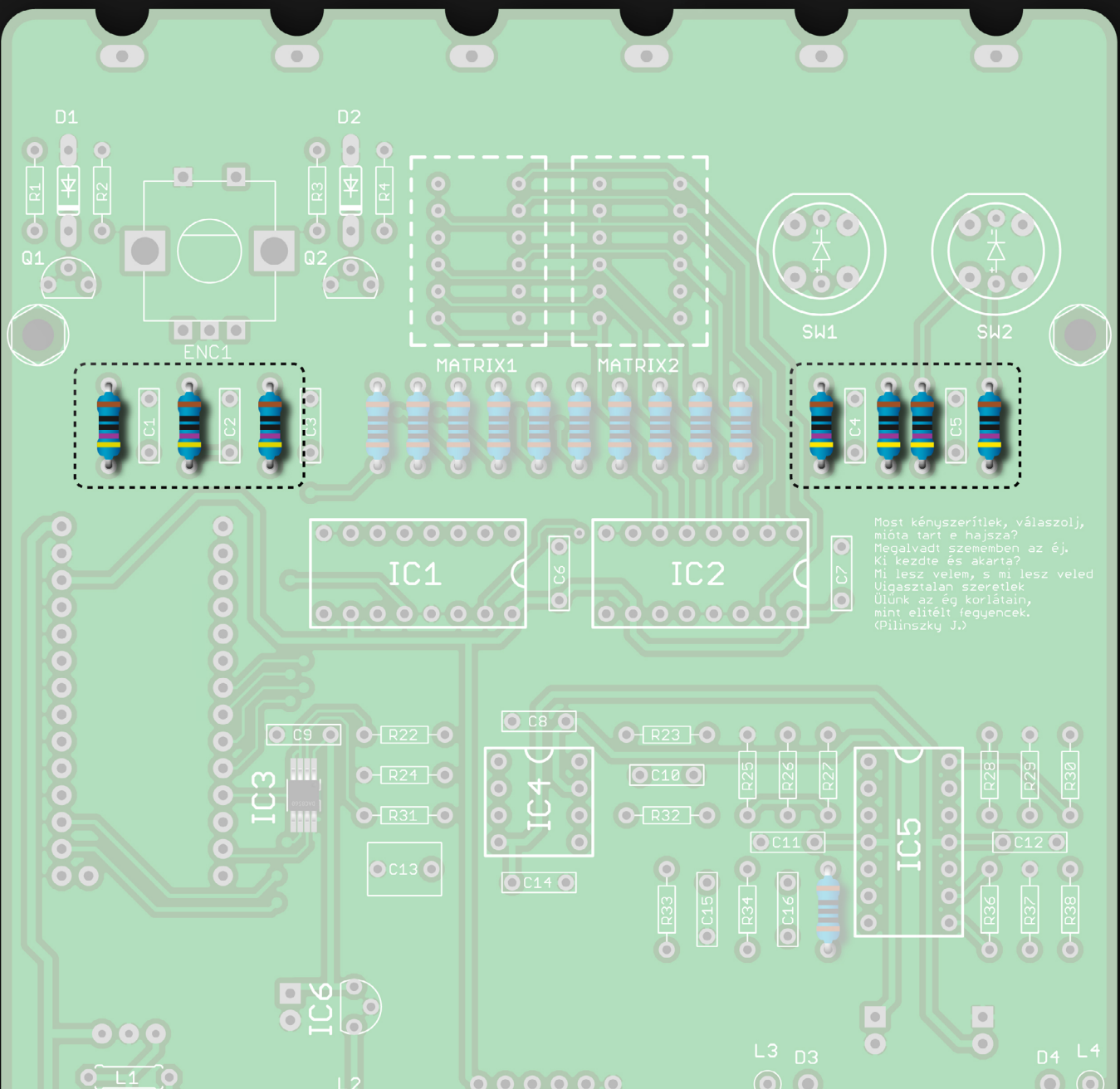


#02 / RESISTORS

11x 100Ω

R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R35

ORIENTATION **DOES NOT MATTER**

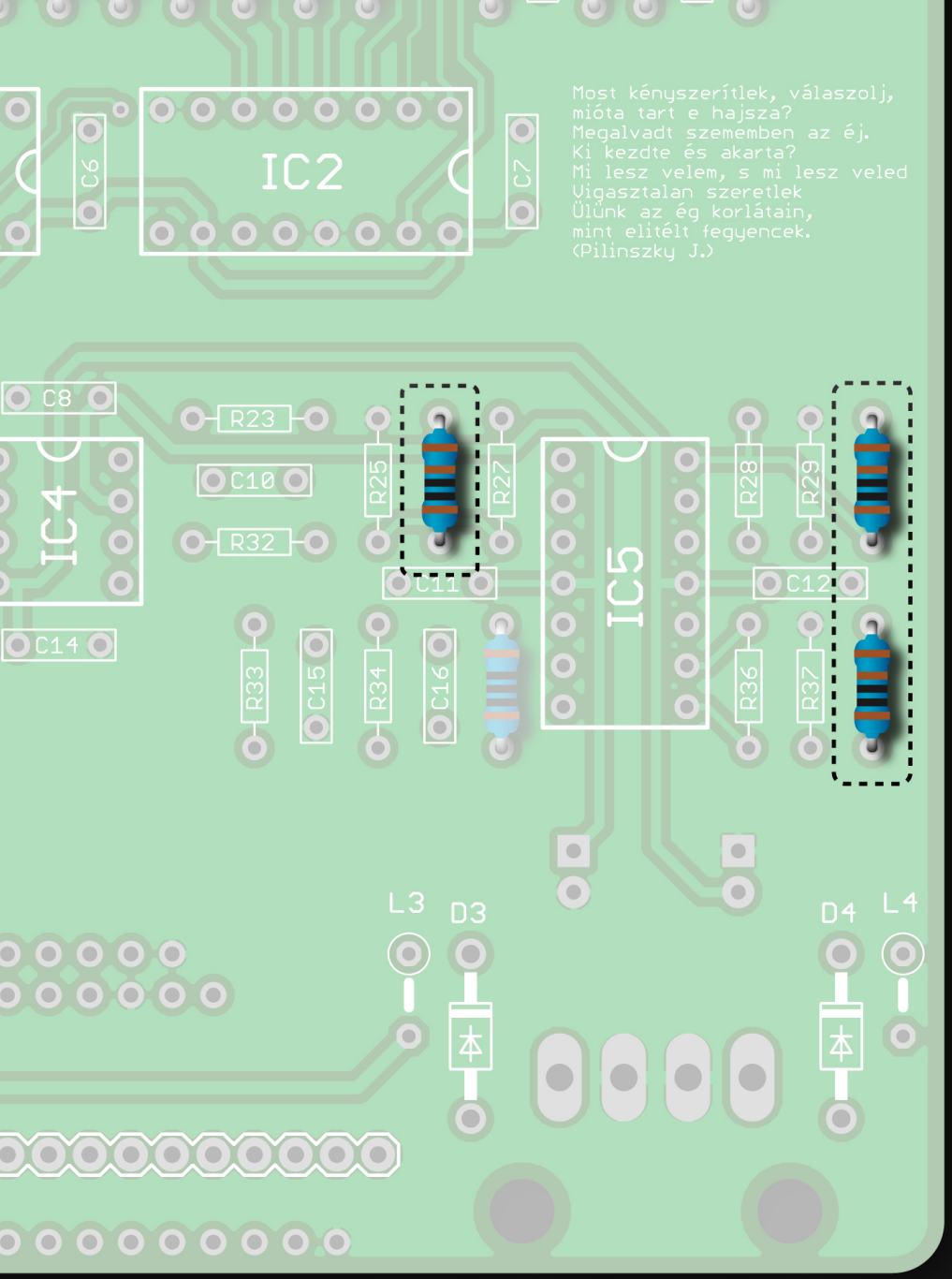


#03 / RESISTORS

7x 470Ω

R5, R6, R7,
R18, R19, R20, R21

ORIENTATION **DOES NOT MATTER**

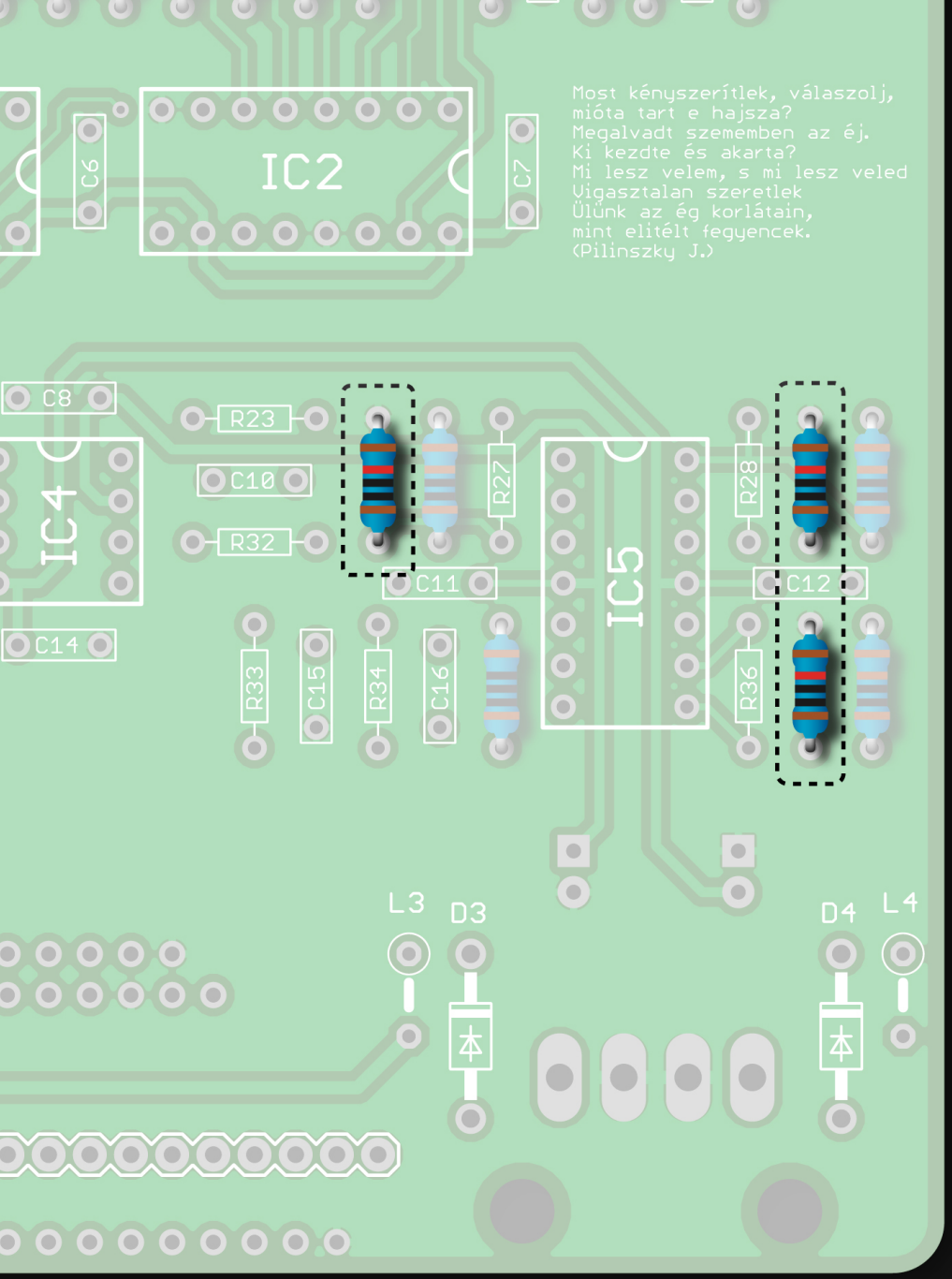


#04 / RESISTORS

3x 1kΩ

R26, R30, R38

ORIENTATION **DOES NOT MATTER**

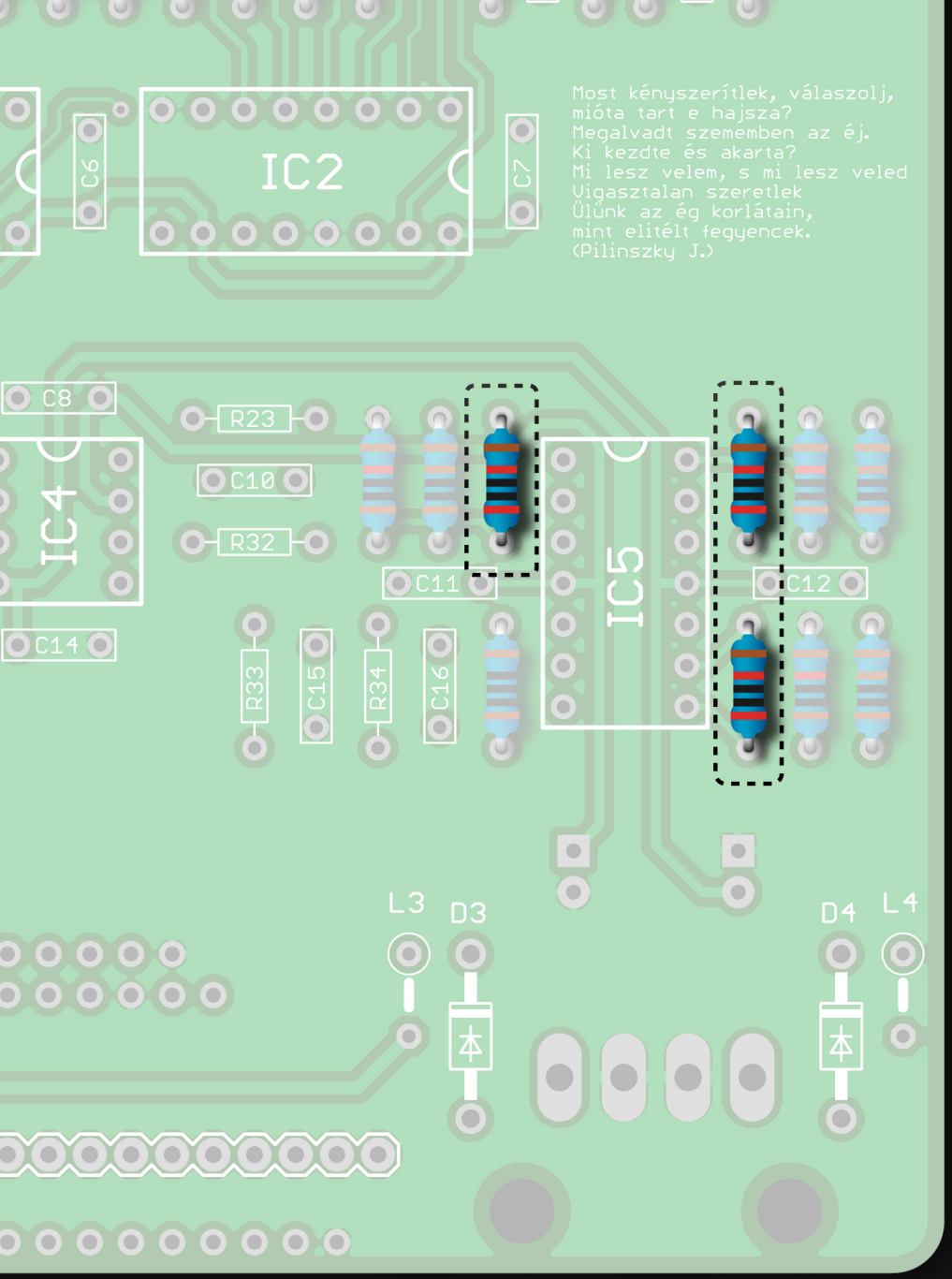


#05 / RESISTORS

3x 10kΩ

R25, R29, R37

ORIENTATION **DOES NOT MATTER**

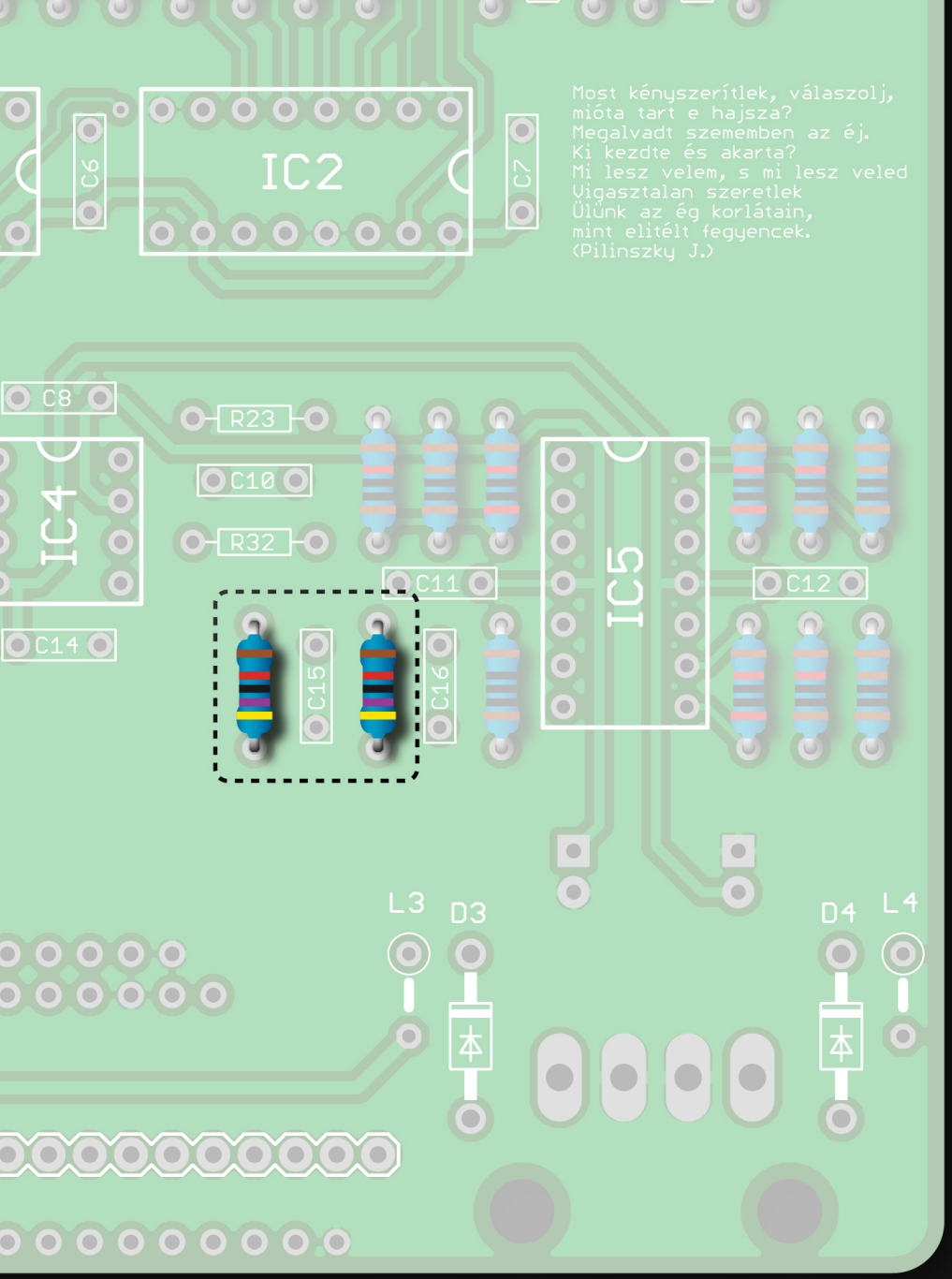


#06 / RESISTORS

3x 20kΩ

R27, R28, R36

ORIENTATION **DOES NOT MATTER**



#07 / RESISTORS

2x 47kΩ

R33, R34

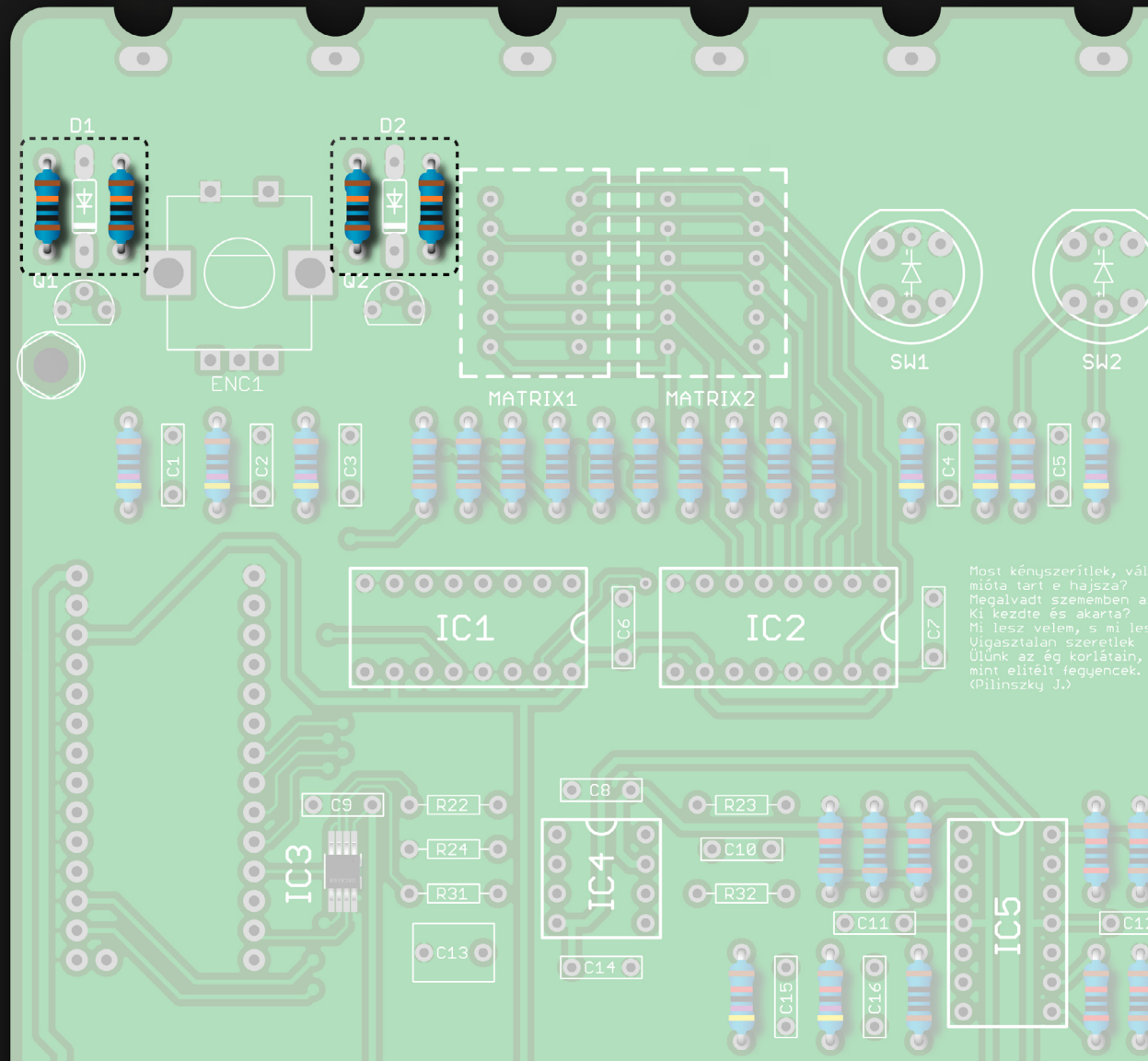
ORIENTATION **DOES NOT MATTER**

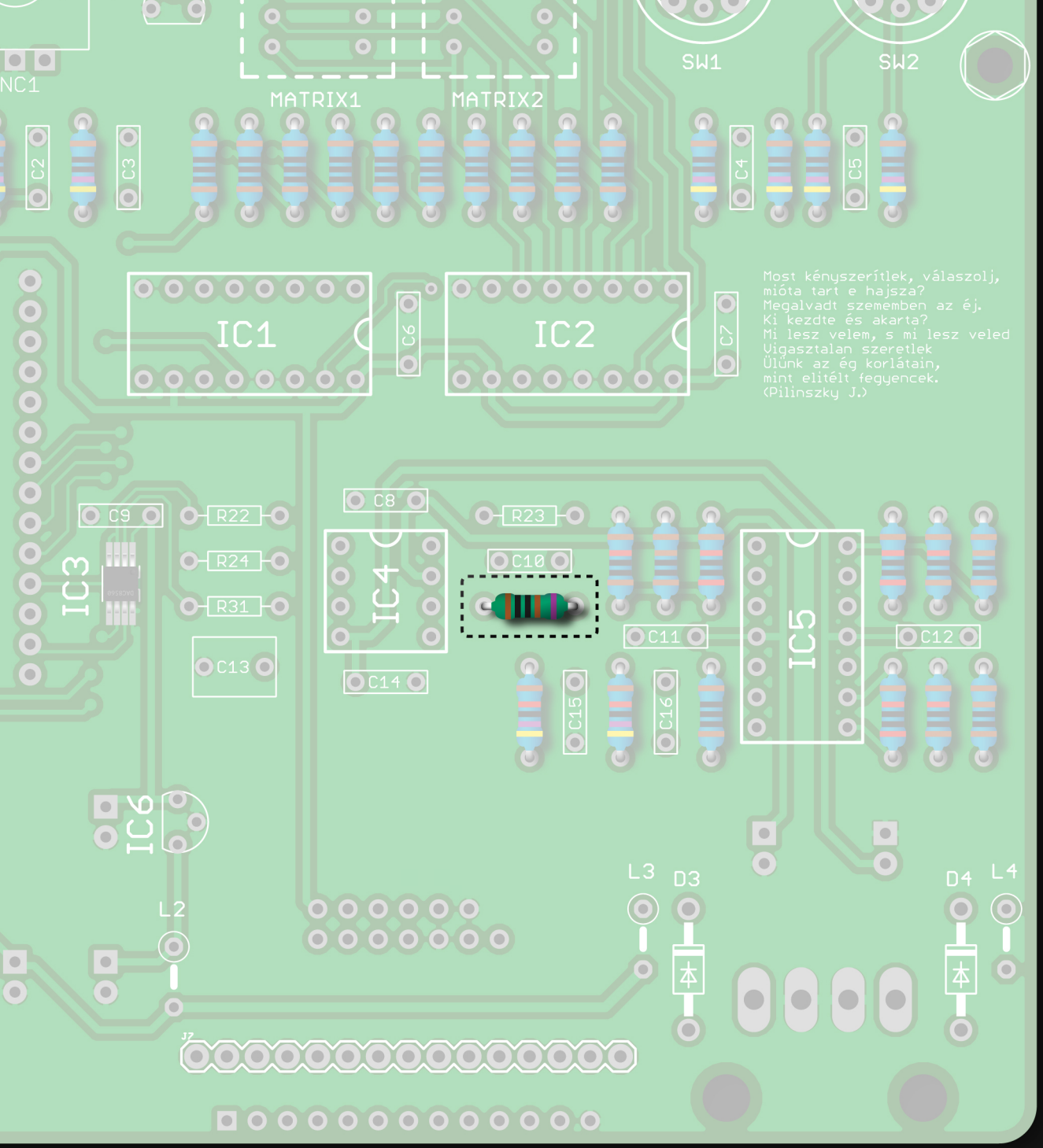
#08 / RESISTORS

4x 100kΩ

R1, R2, R3, R4

ORIENTATION **DOES NOT MATTER**



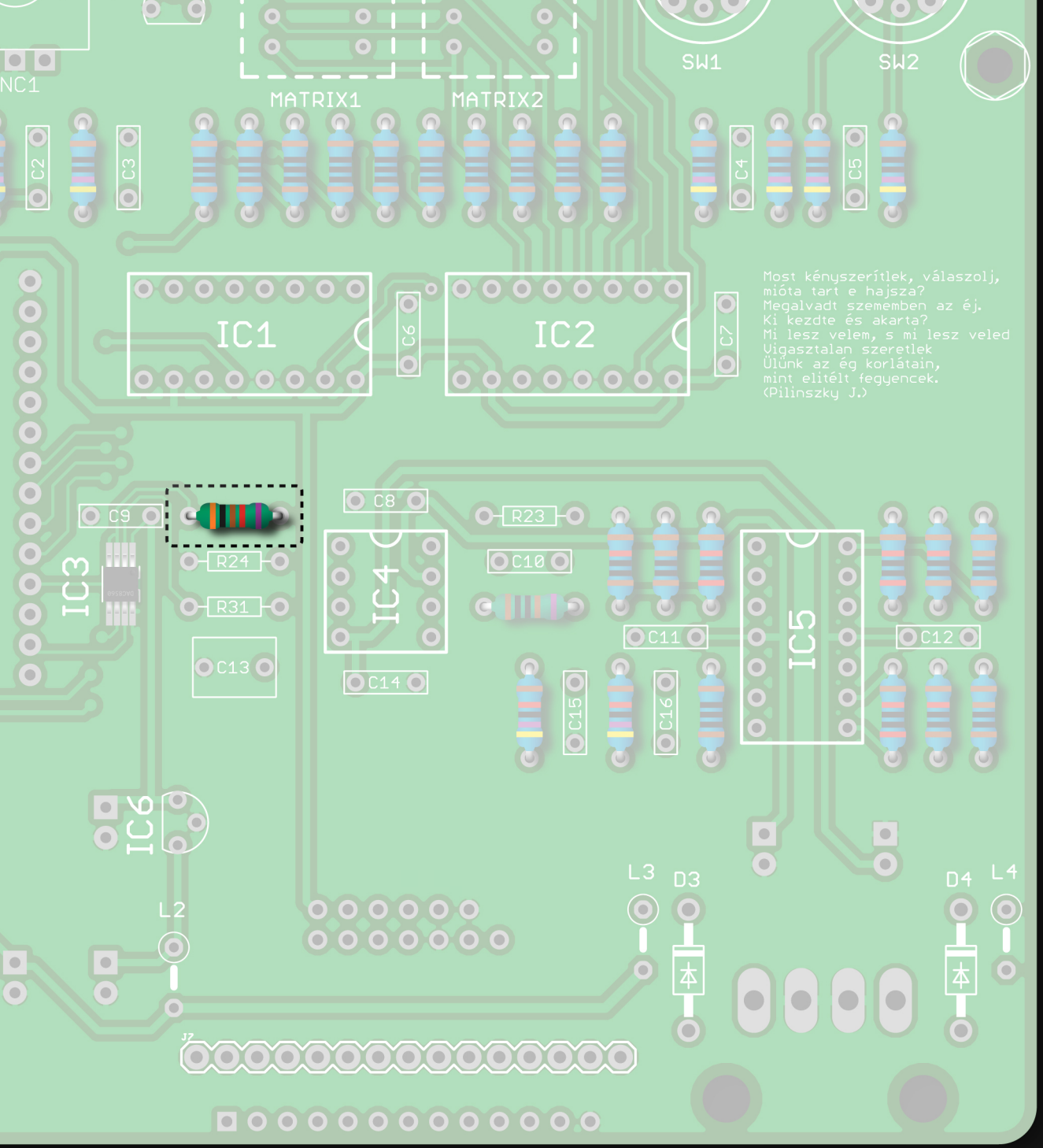


#09 / PRECISION RESISTORS

1x 1kΩ (0.1%)

R32

ORIENTATION **DOES NOT MATTER**

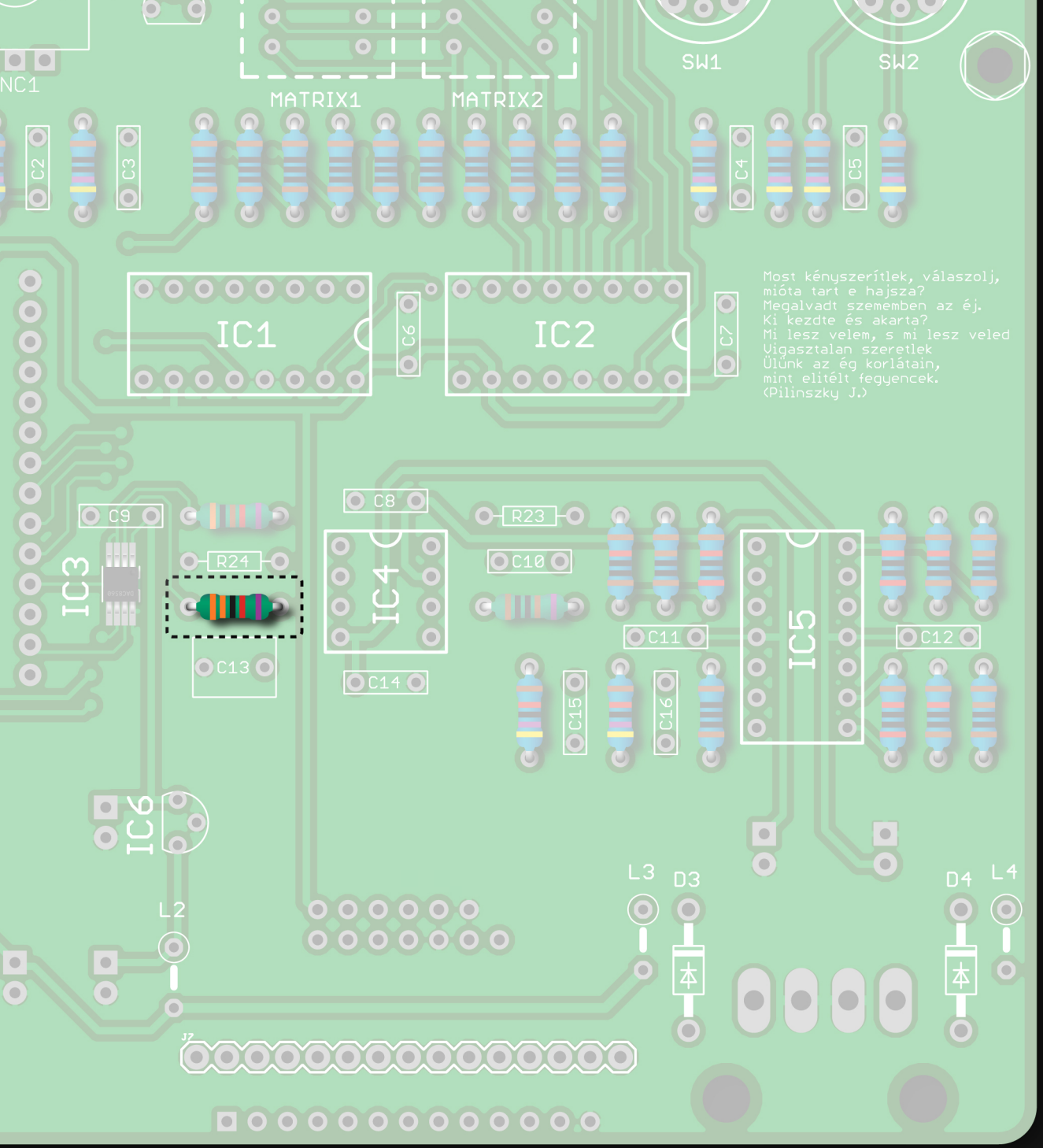


#10 / PRECISION RESISTORS

1x 30.1kΩ (0.1%)

R22

ORIENTATION **DOES NOT MATTER**

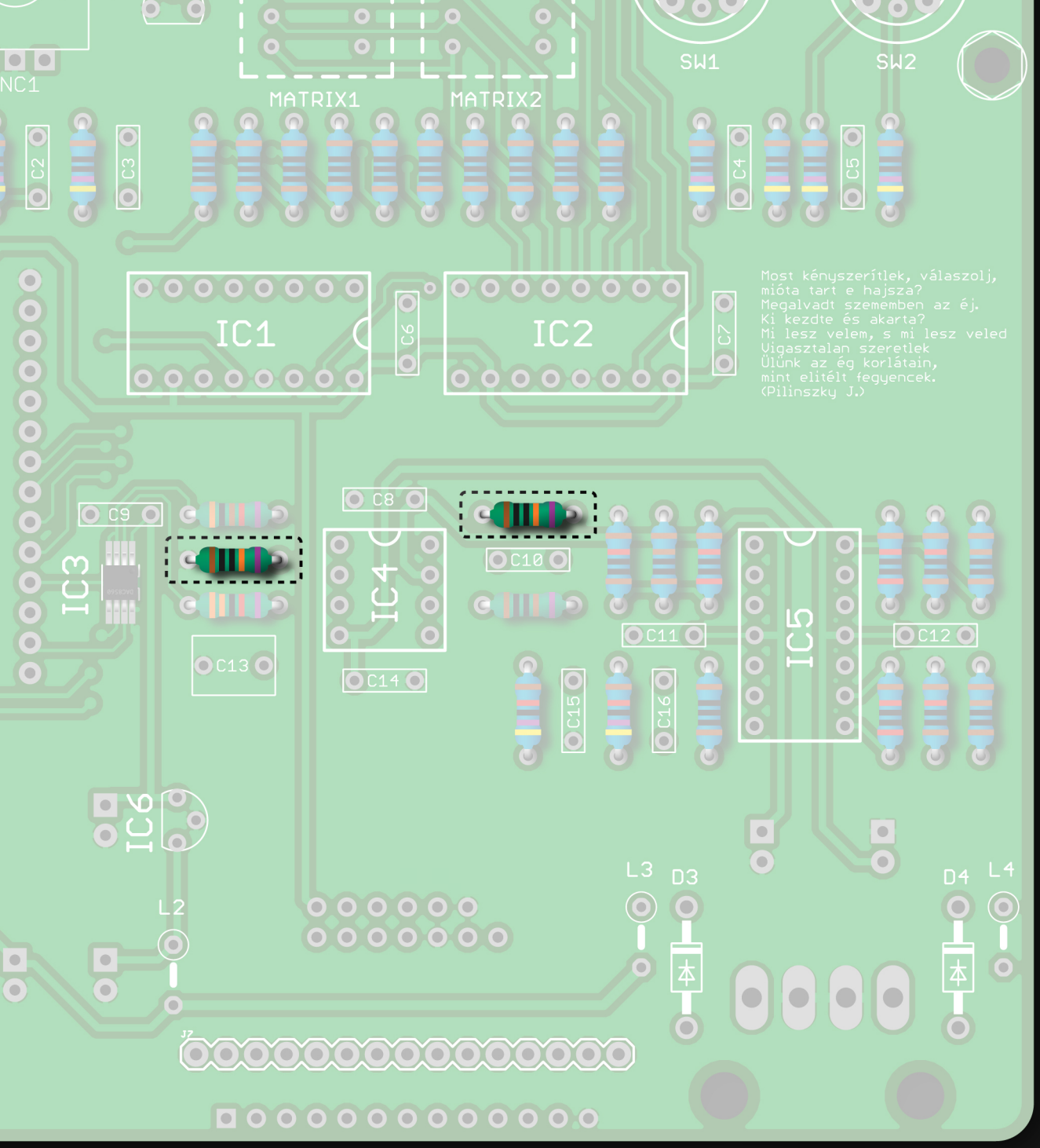


#11 / PRECISION RESISTORS

1x 33kΩ (0.1%)

R31

ORIENTATION **DOES NOT MATTER**



#12 / PRECISION RESISTORS

2x 100kΩ (0.1%)

R23, R24

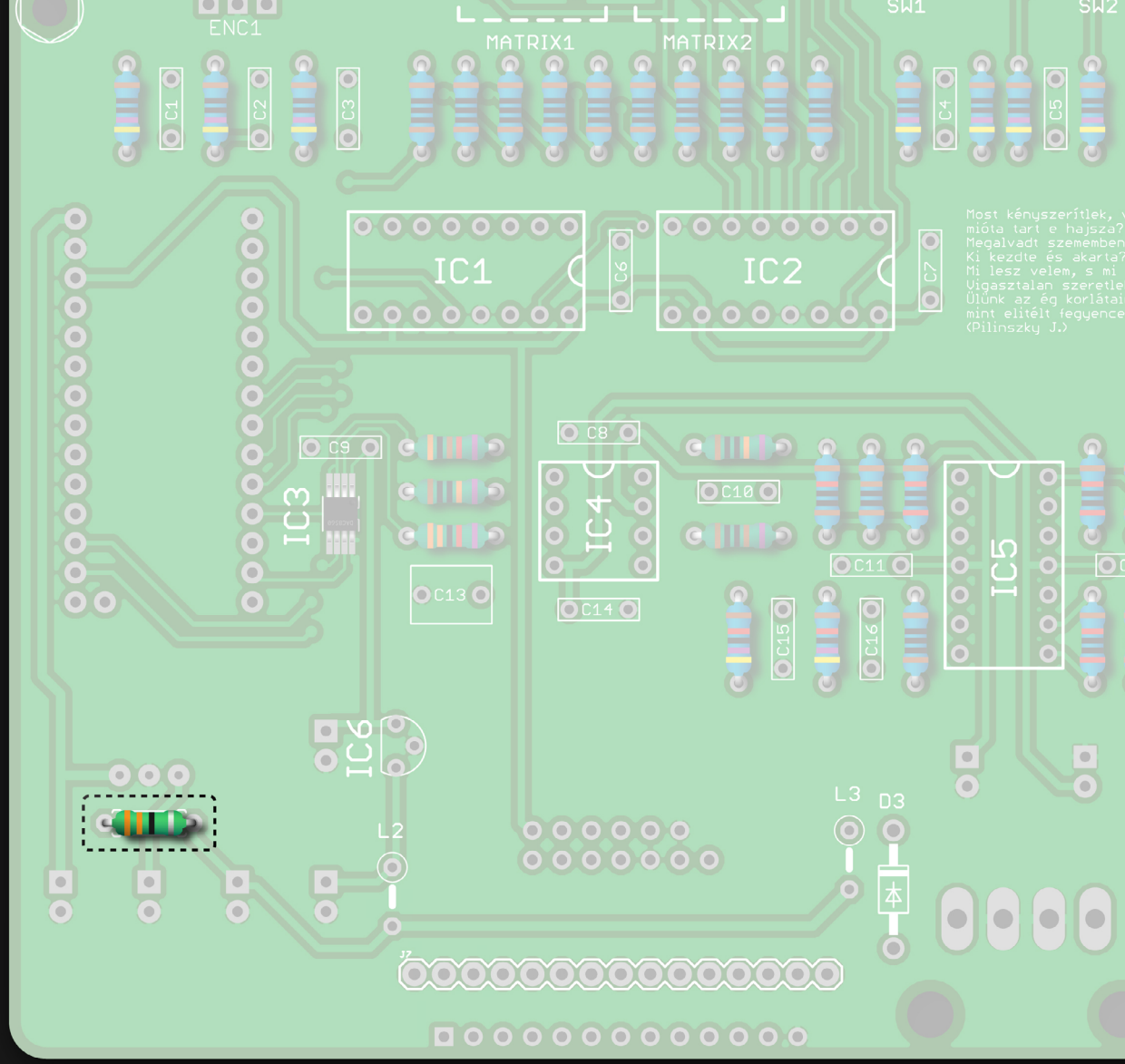
ORIENTATION **DOES NOT MATTER**

#13 / INDUCTOR

1x 33µH

L1

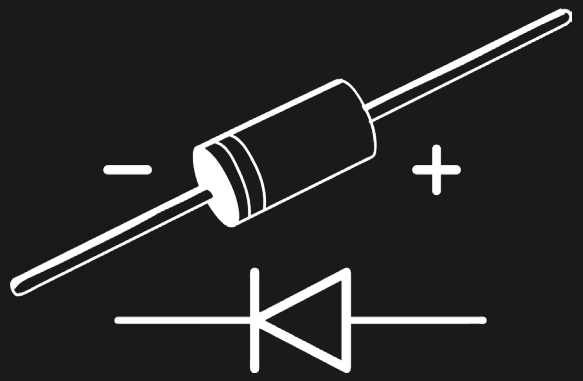
ORIENTATION **DOES NOT MATTER**

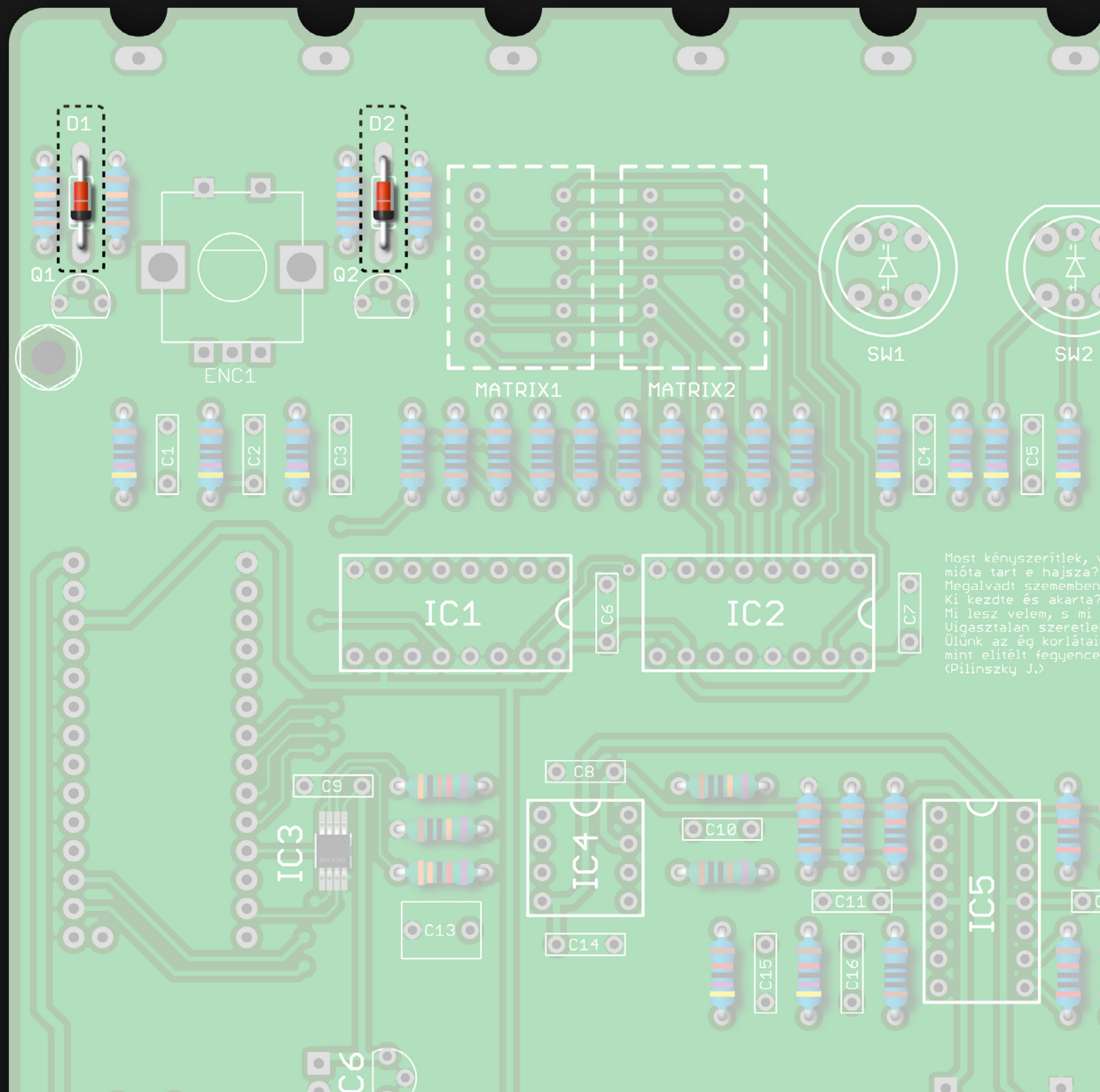


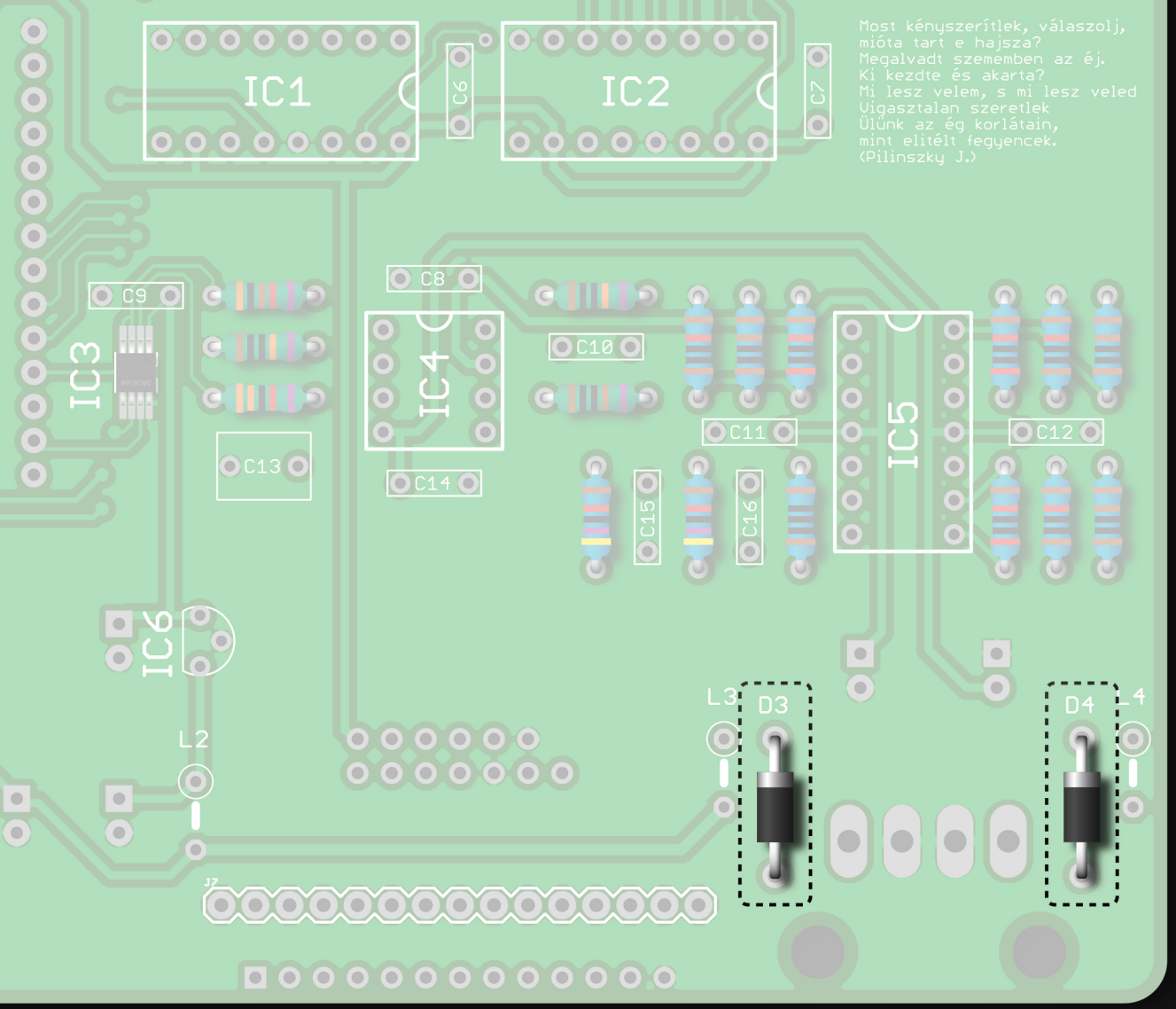
#14 / DIODES

2x 1N4148
D1, D2

ORIENTATION: THE BLACK MARKING (SYMBOLIZING THE CATHODE OF THE DIODE) SHOULD BE ON THE BOTTOM



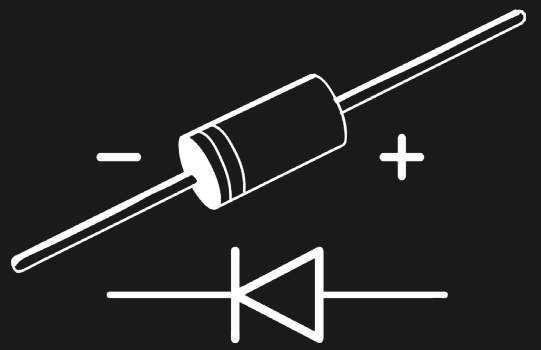


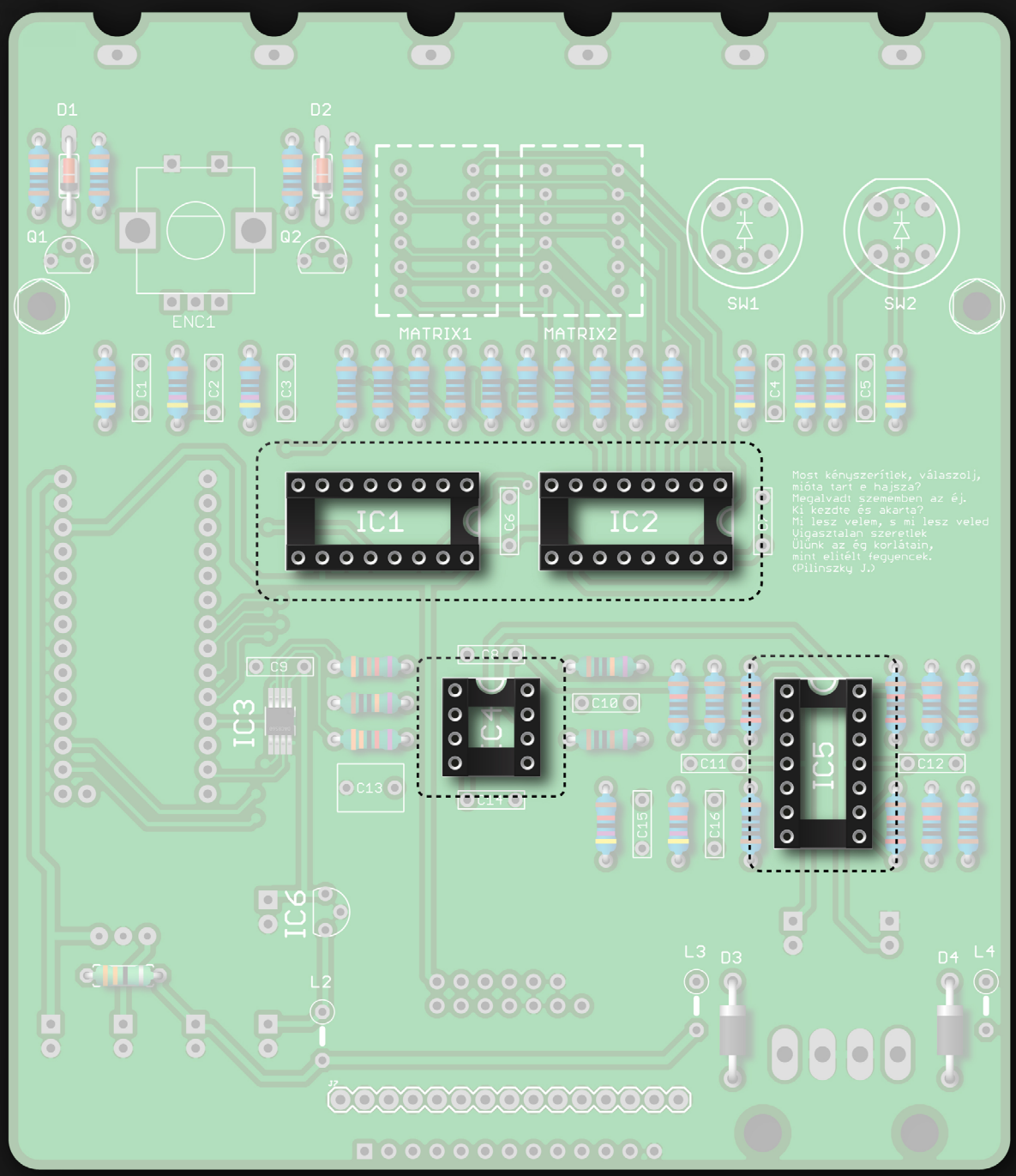


#15 / DIODES

2x 1N5819
D3, D4

ORIENTATION: **THE WHITE MARKING (SYMBOLIZING THE CATHODE OF THE DIODE) SHOULD BE ON THE TOP**





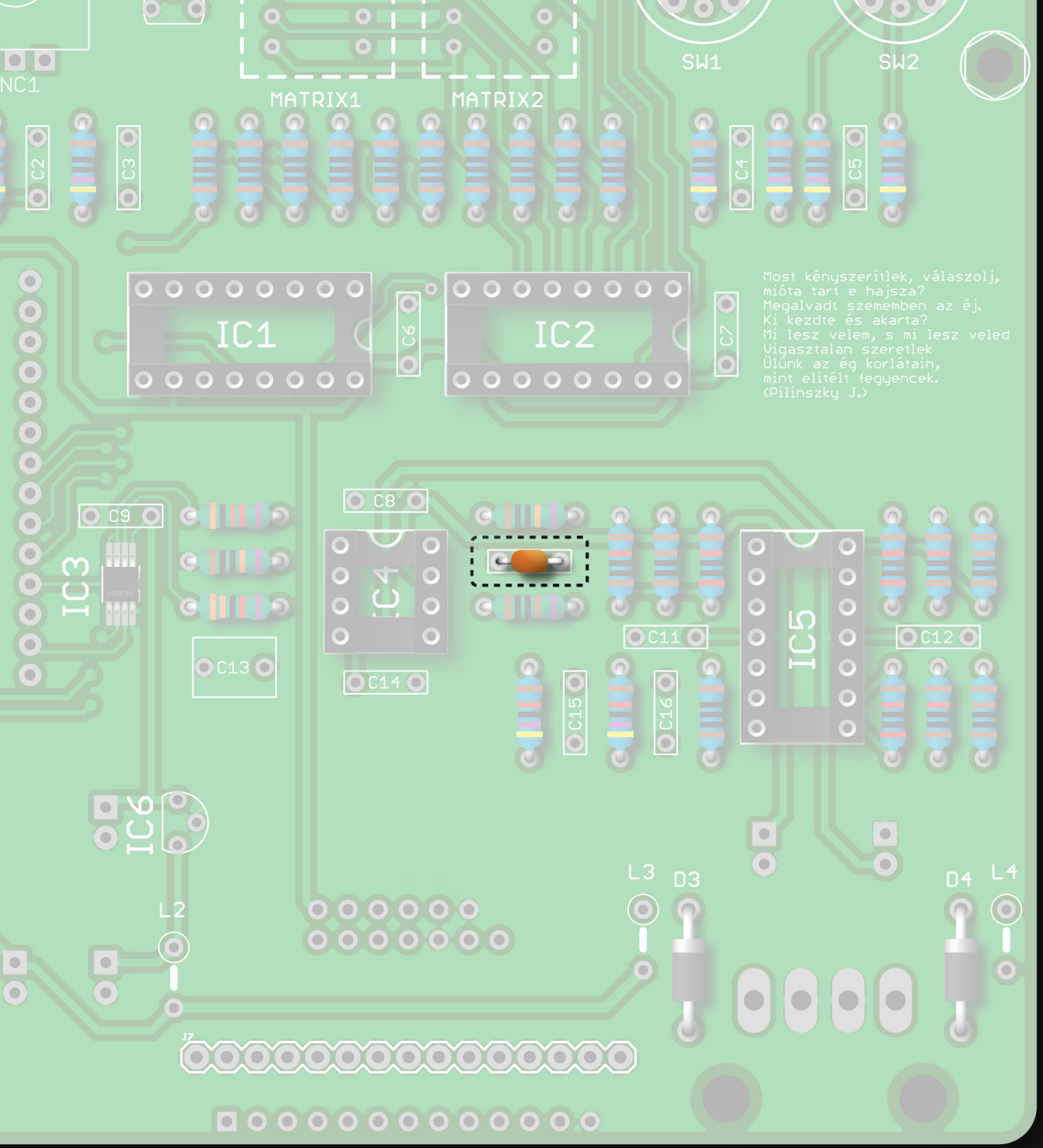
#16 / IC SOCKETS

2x 16-PIN IC SOCKET IC1, IC2

ORIENTATION: THE NOTCH SHOULD BE ON THE RIGHT

1x 8-PIN IC SOCKET IC4
1x 14-PIN IC SOCKET IC5

ORIENTATION: THE NOTCH SHOULD BE ON THE TOP

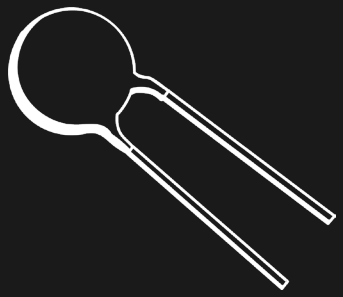


#17 / CAPACITORS

1x 22pF

C10

ORIENTATION **DOES NOT MATTER**

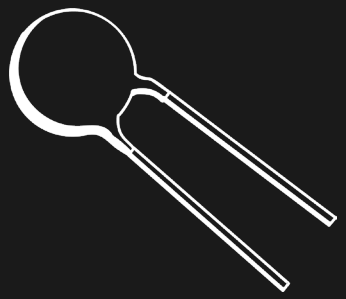


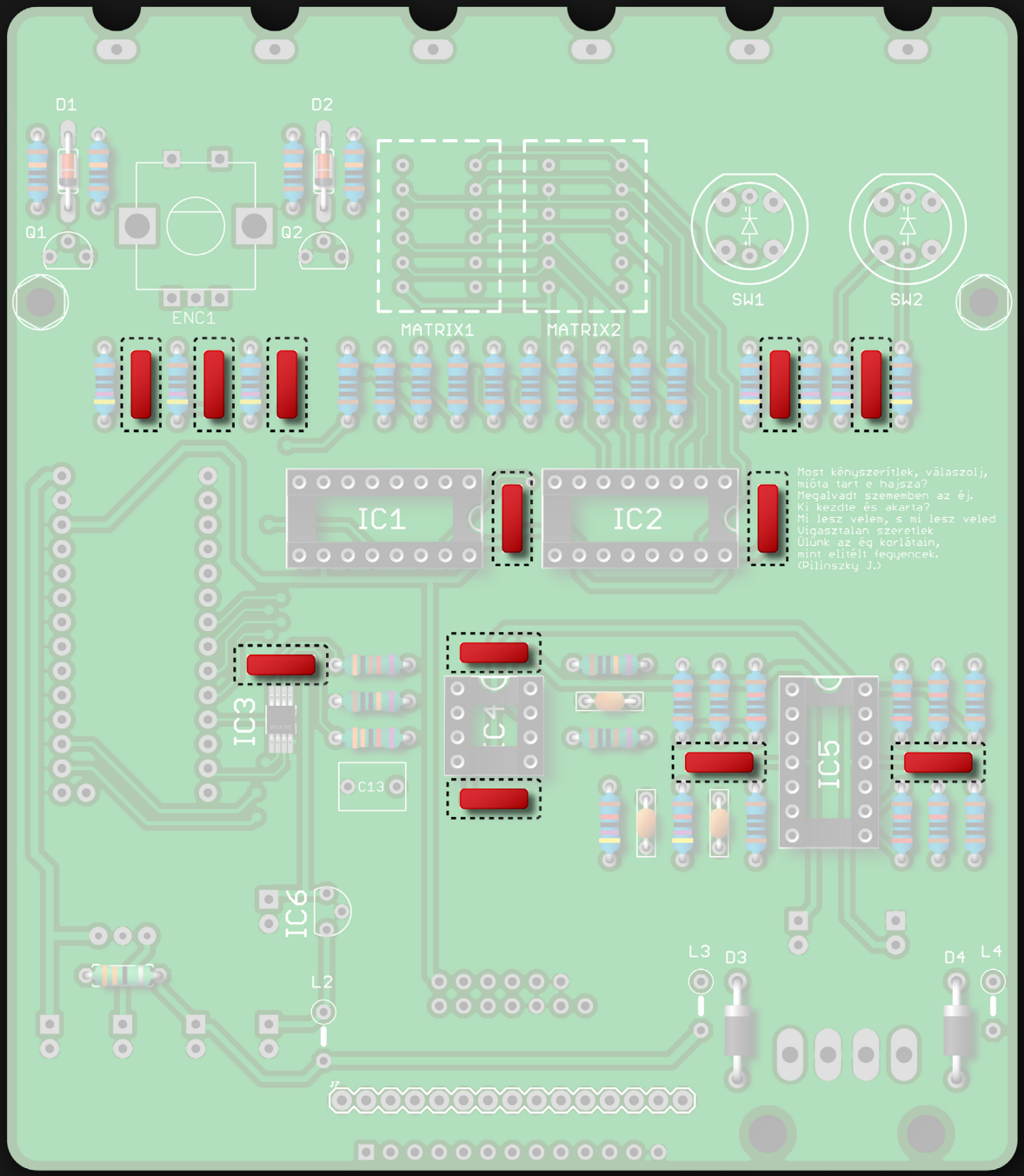
#18 / CAPACITORS

2x 220pF

C15, C16

ORIENTATION **DOES NOT MATTER**



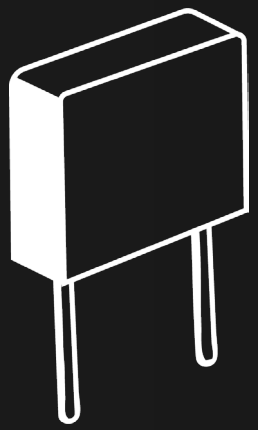


#19 / CAPACITORS

12x 100nF

C1, C2, C3, C4, C5, C6, C7, C8, C9, C11, C12, C14

ORIENTATION **DOES NOT MATTER**

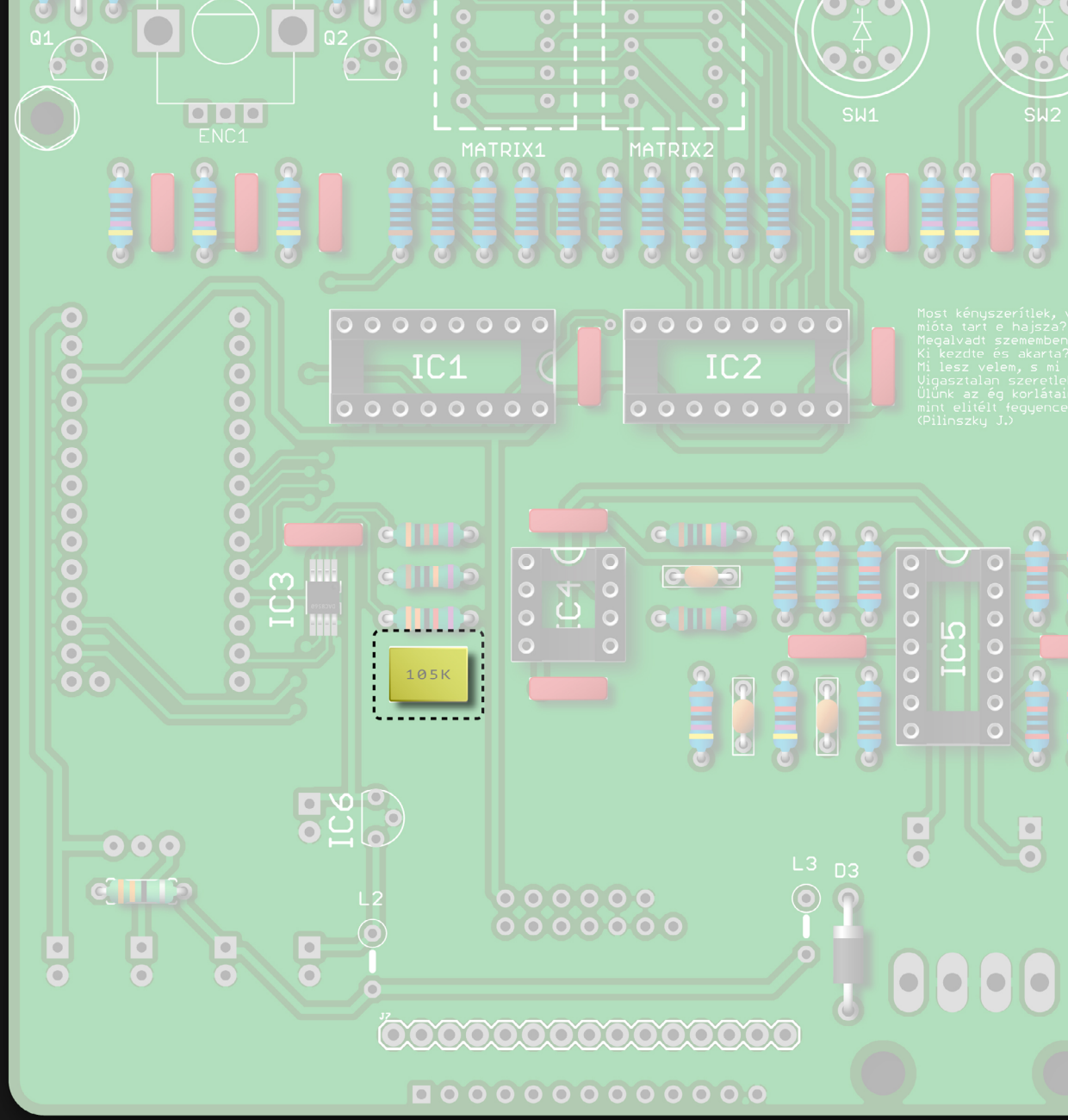


#20 / CAPACITORS

1x 1µF

C13

ORIENTATION **DOES NOT MATTER**

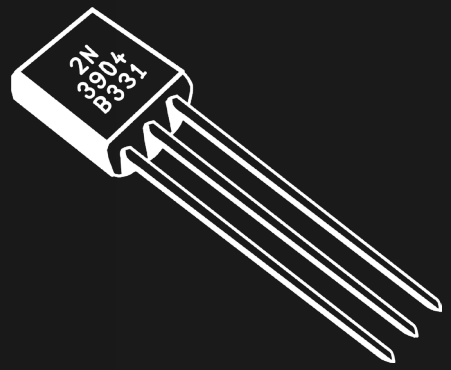


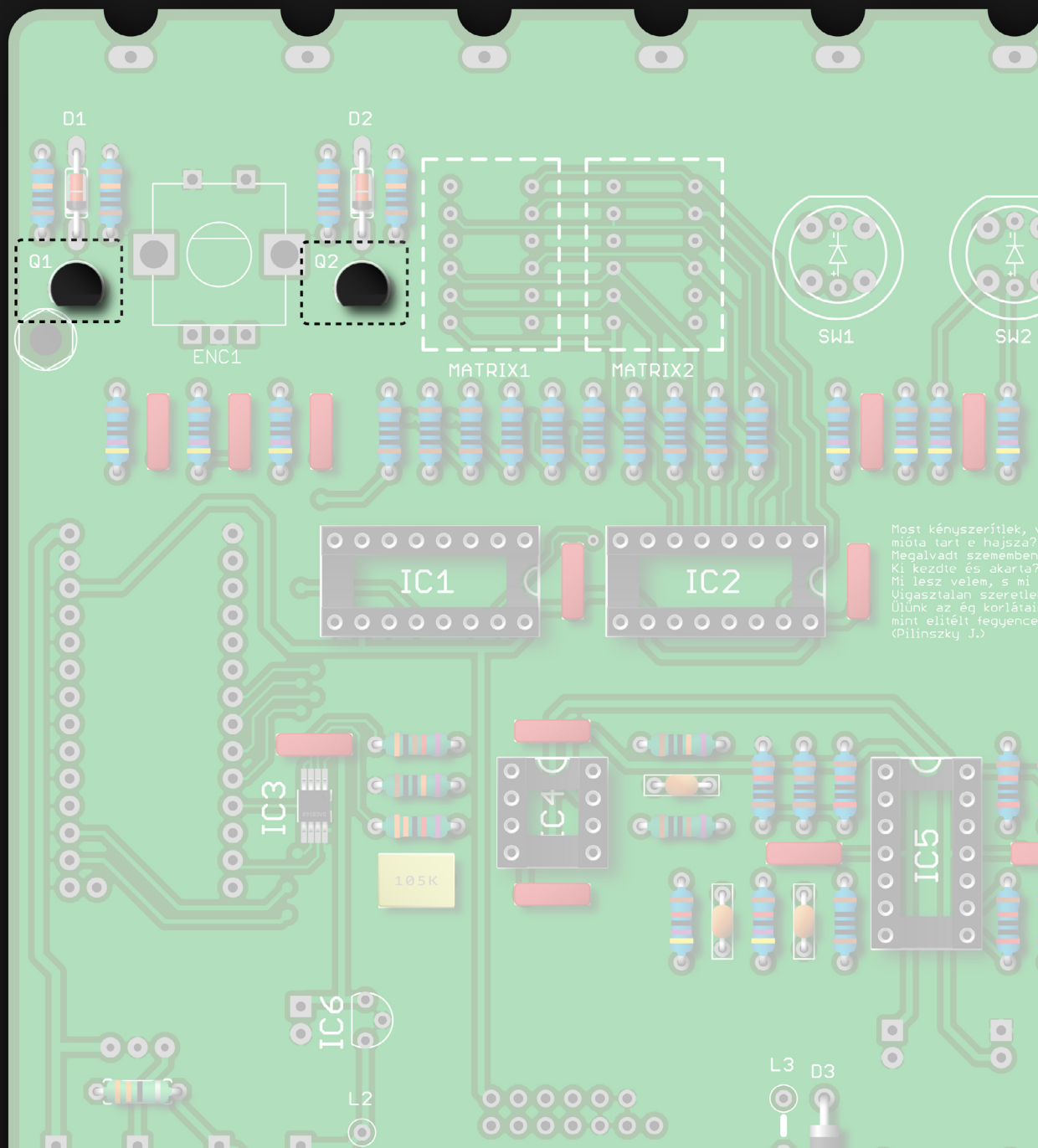
#21 / TRANSISTORS

2x 2N3904

Q1, Q2

ORIENTATION: **THE FLAT SIDE OF TRANSISTORS SHOULD BE ON THE BOTTOM, MATCH UP WITH THE SILKSCREEN MARKING**



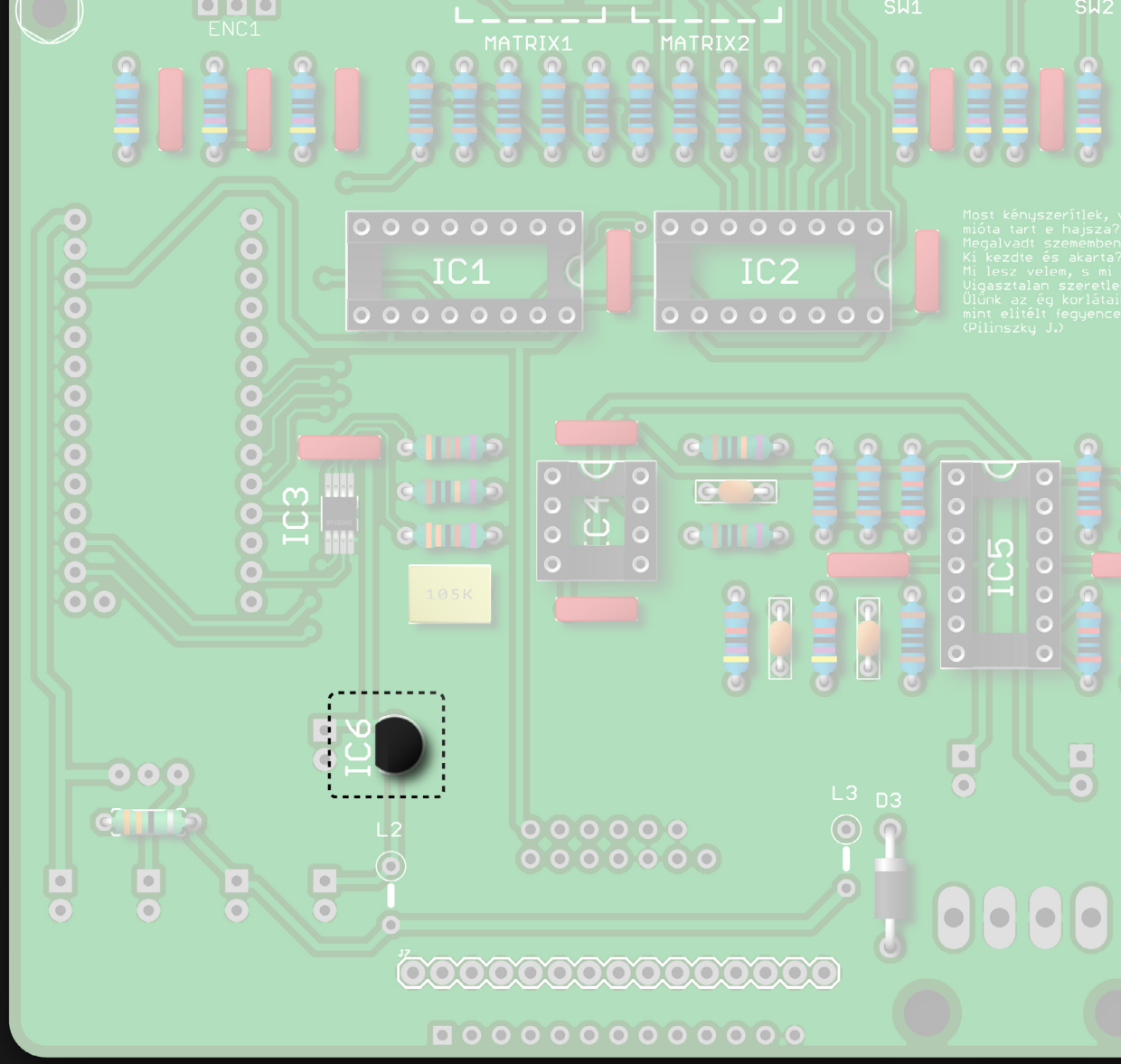


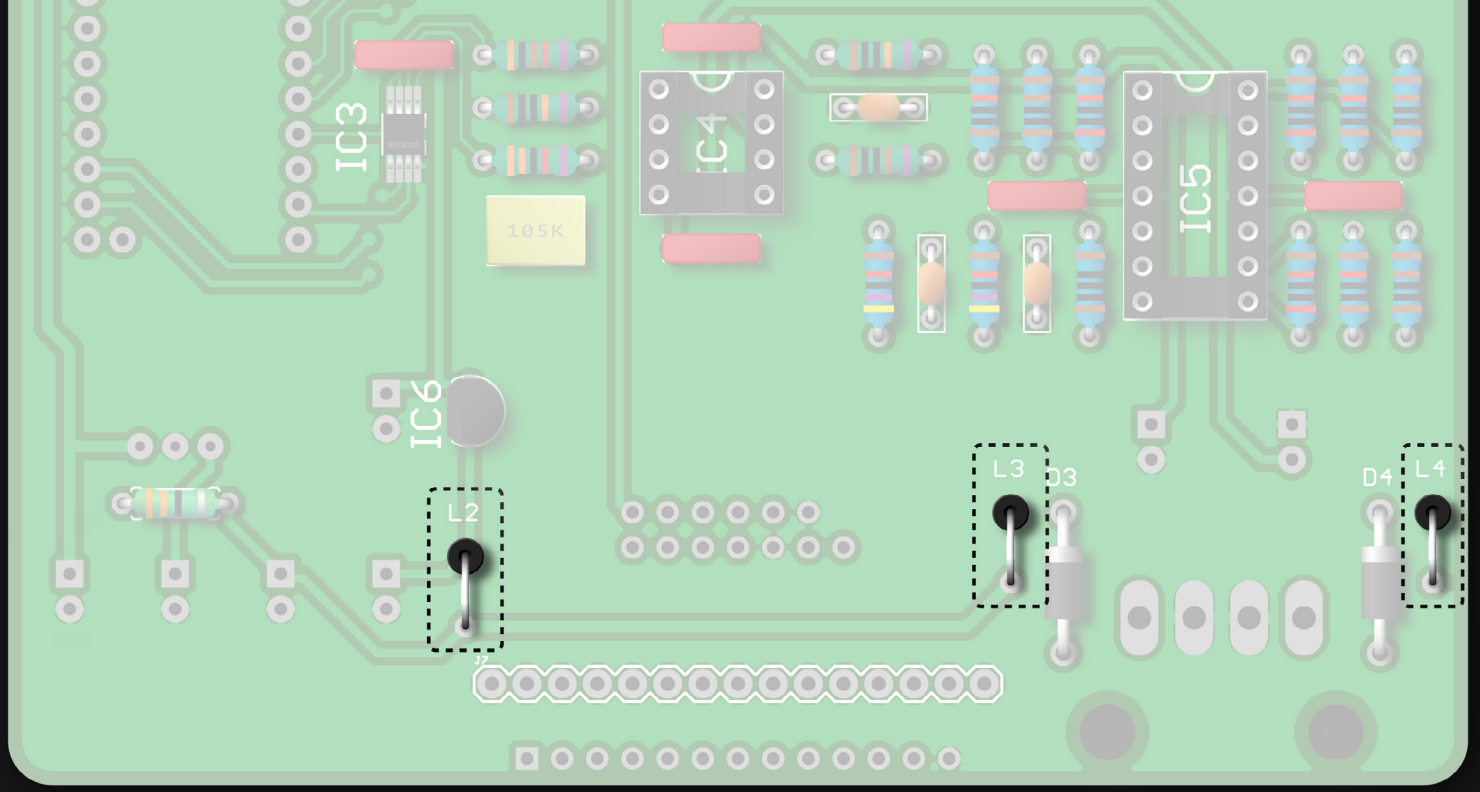
#22 / VOLTAGE REGULATOR

1x L78L33

IC6

ORIENTATION: THE FLAT SIDE OF THE REGULATOR SHOULD BE ON THE LEFT, MATCH UP WITH THE SILKSCREEN MARKING



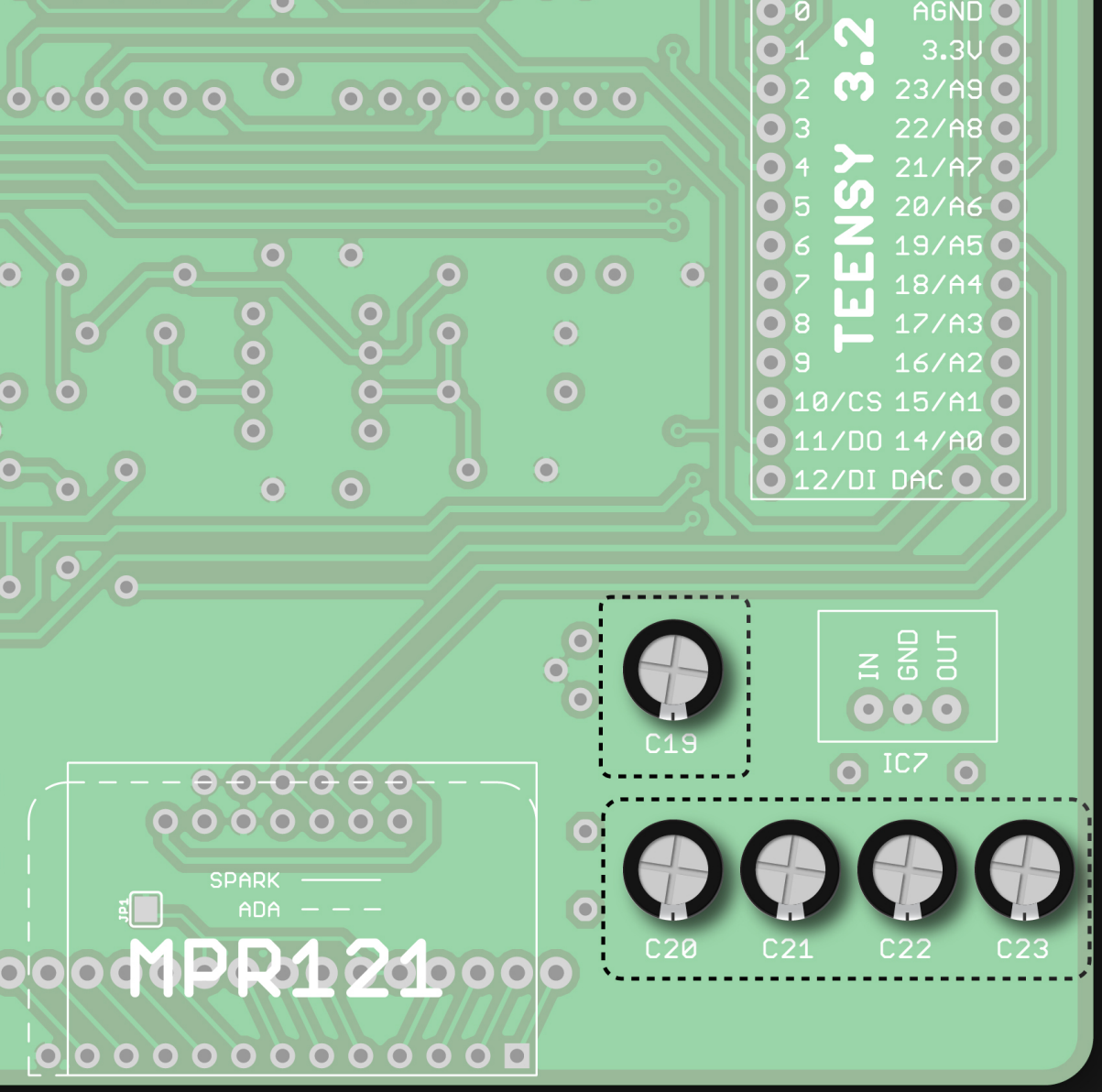


#23 / FERRITE BEADS

3x BLO2RN1R2M2B

L2, L3, L4

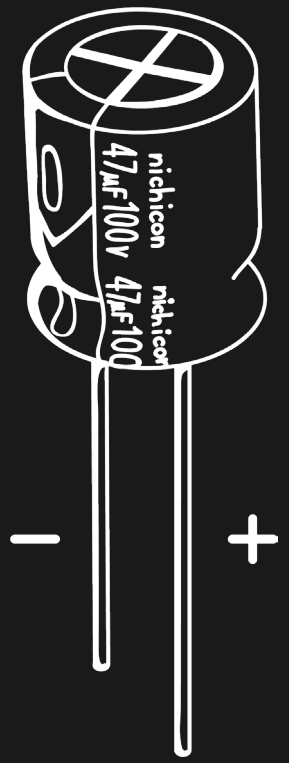
ORIENTATION: **MATCH UP WITH THE SILKSCREEN MARKING**



#24 / ELECTROLYTIC CAPACITORS

5x 22µF C19, C20, C21, C22, C23

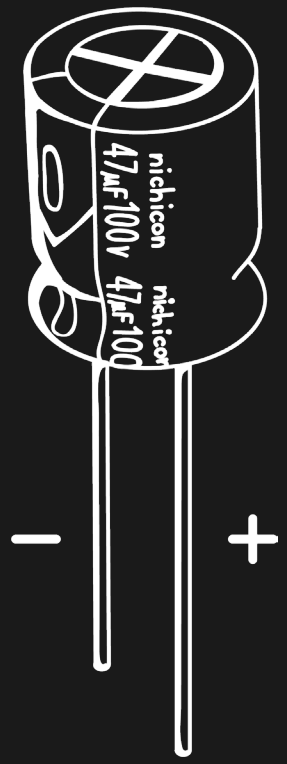
ORIENTATION: THE LONGER LEADS GOES TO HOLES MARKED WITH “+” , “- - -” MARKING SHOULD FACE THE BOTTOM OF THE PCB, AS PICTURED

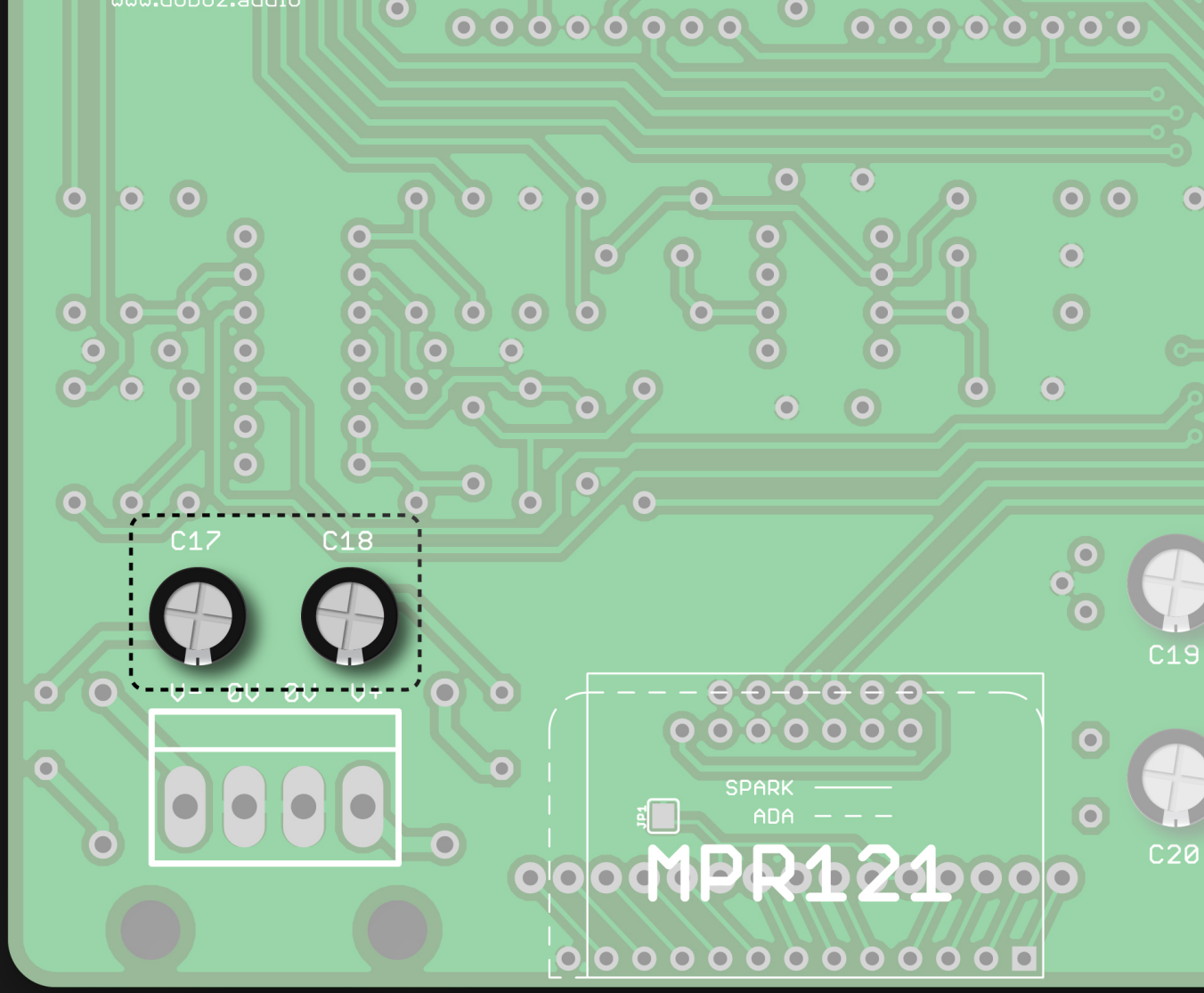


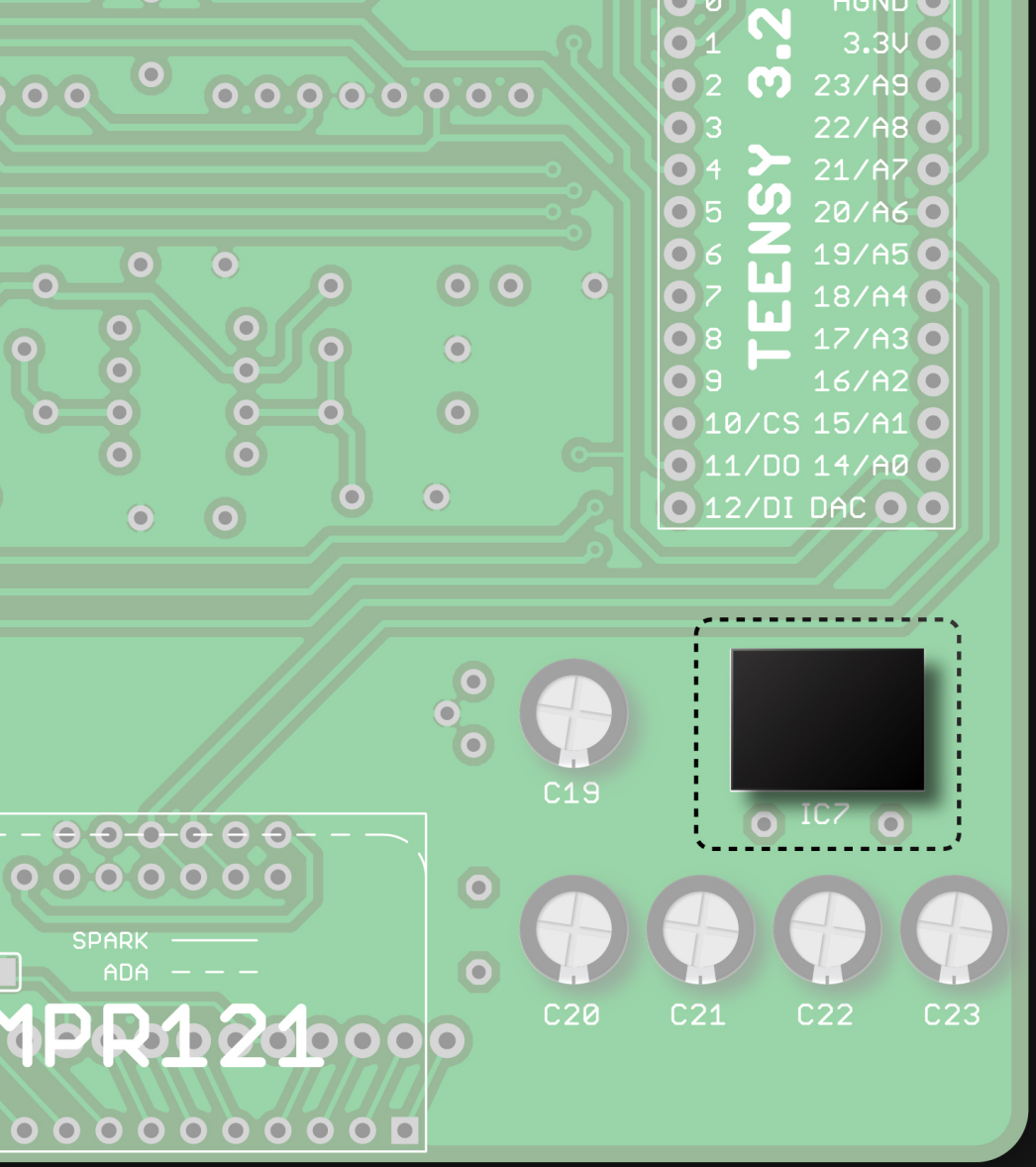
#25 / ELECTROLYTIC CAPACITORS

2x 47µF
C17, C18

ORIENTATION: THE LONGER LEADS GOES TO HOLES MARKED WITH “+” , “- - -” MARKING SHOULD FACE THE BOTTOM OF THE PCB, AS PICTURED







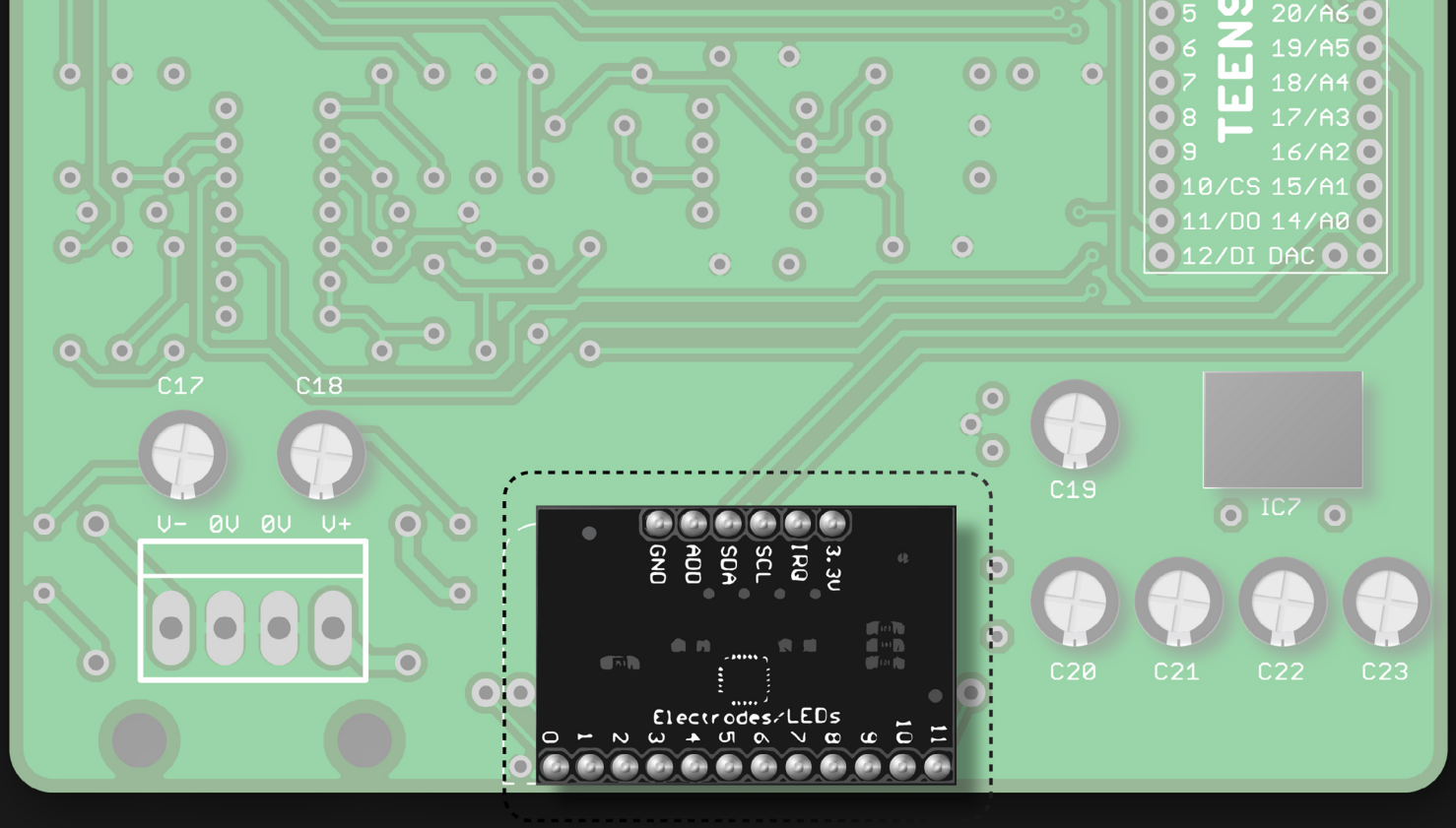
#26 / MISC. COMPONENTS

5V DC/DC CONVERTER*

IC7

ORIENTATION: **SOLDER IT TO THE BACKSIDE OF THE PCB, MATCH UP WITH THE SILKSCREEN MARKING!**

* R-78E5.0-0.5 or AMSR-7805-NZ

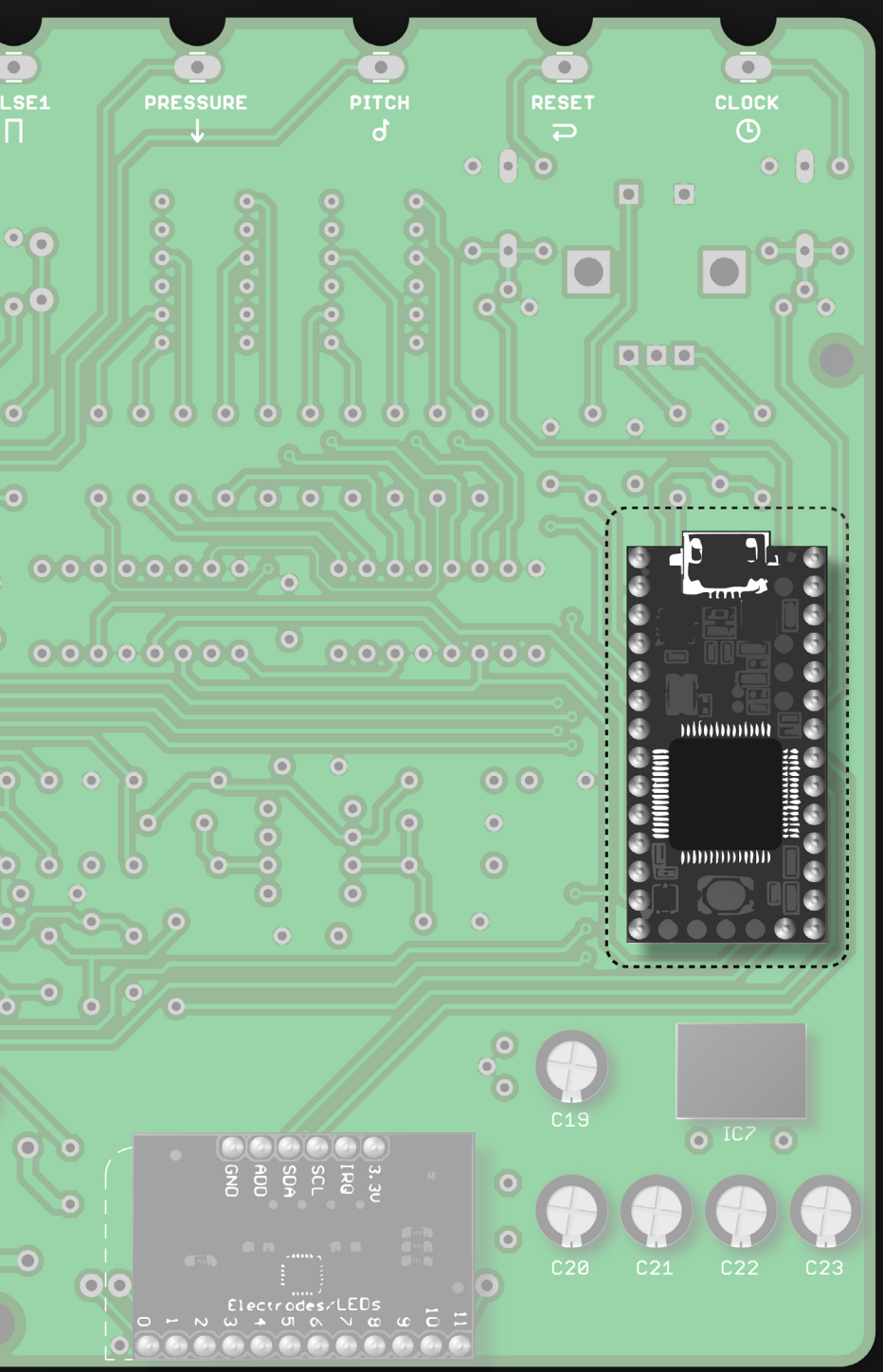


#27 / MISC. COMPONENTS

MPR121

BREAKOUT BOARD

SOLDER IT TO THE BACKSIDE OF THE PCB, MATCH UP WITH THE SILKSCREEN MARKING!
USING BREAKABLE FEMALE HEADERS AS SOCKET IS HIGHLY RECOMMENDED!
TSNM MKII ACCEPTS THE CHEAPER CHINESE MADE (SPARKFUN CLONE) MPR121 BREAKOUT BOARDS,
AS WELL AS THE BLUE ONES MADE BY ADAFRUIT (THEIR DIMENSION AND PINOUT IS SLIGHTLY DIFFERENT).



#28 / MISC. COMPONENTS

TEENSY 3.2

MICROCONTROLLER BOARD

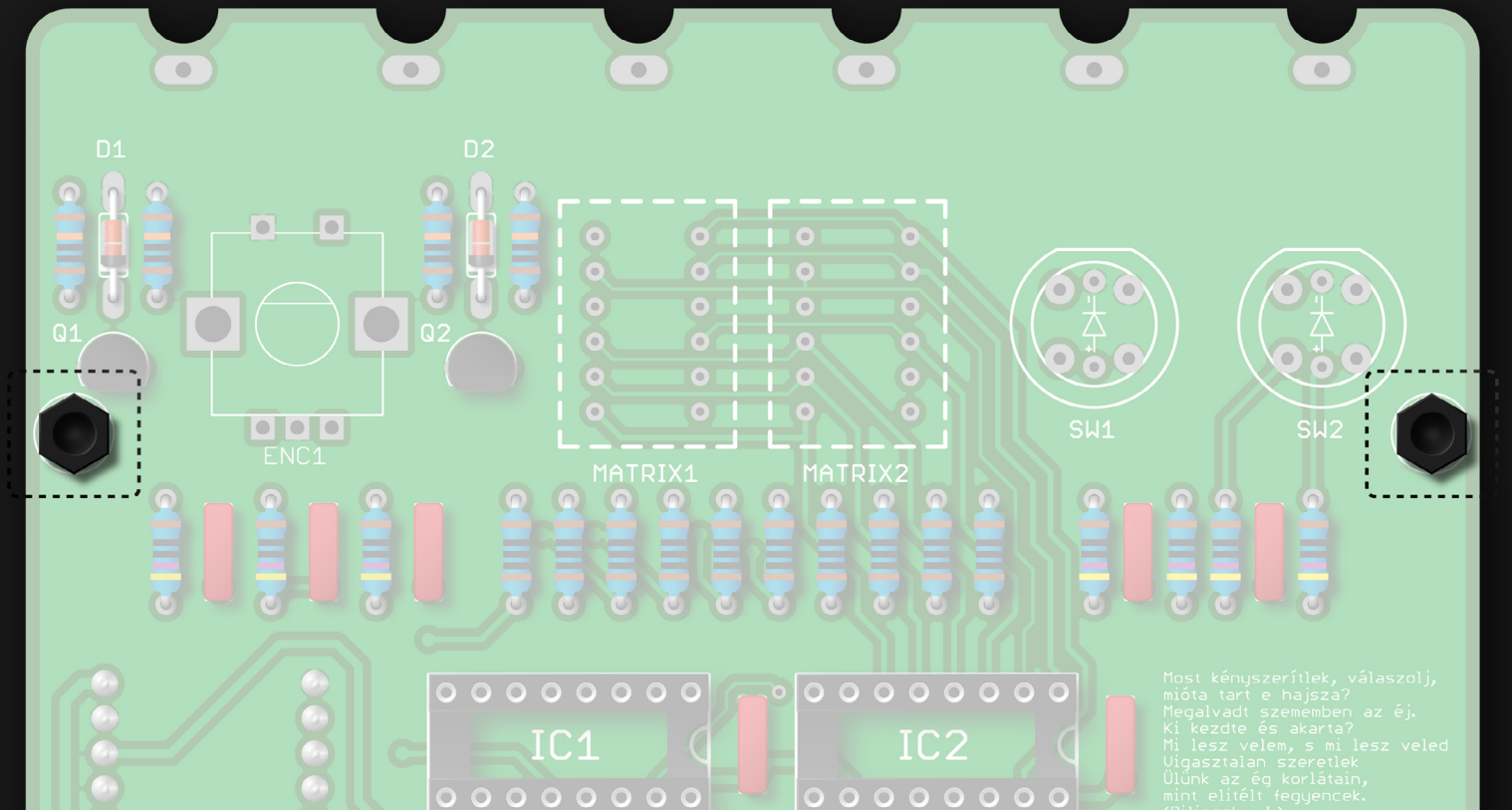
SOLDER IT TO THE BACKSIDE OF THE PCB, MATCH UP WITH THE SILKSCREEN MARKING. USING BREAKABLE FEMALE HEADERS AS SOCKET IS HIGHLY RECOMMENDED. THE USB CONNECTOR SHOULD BE ON THE TOP, AS PICTURED.

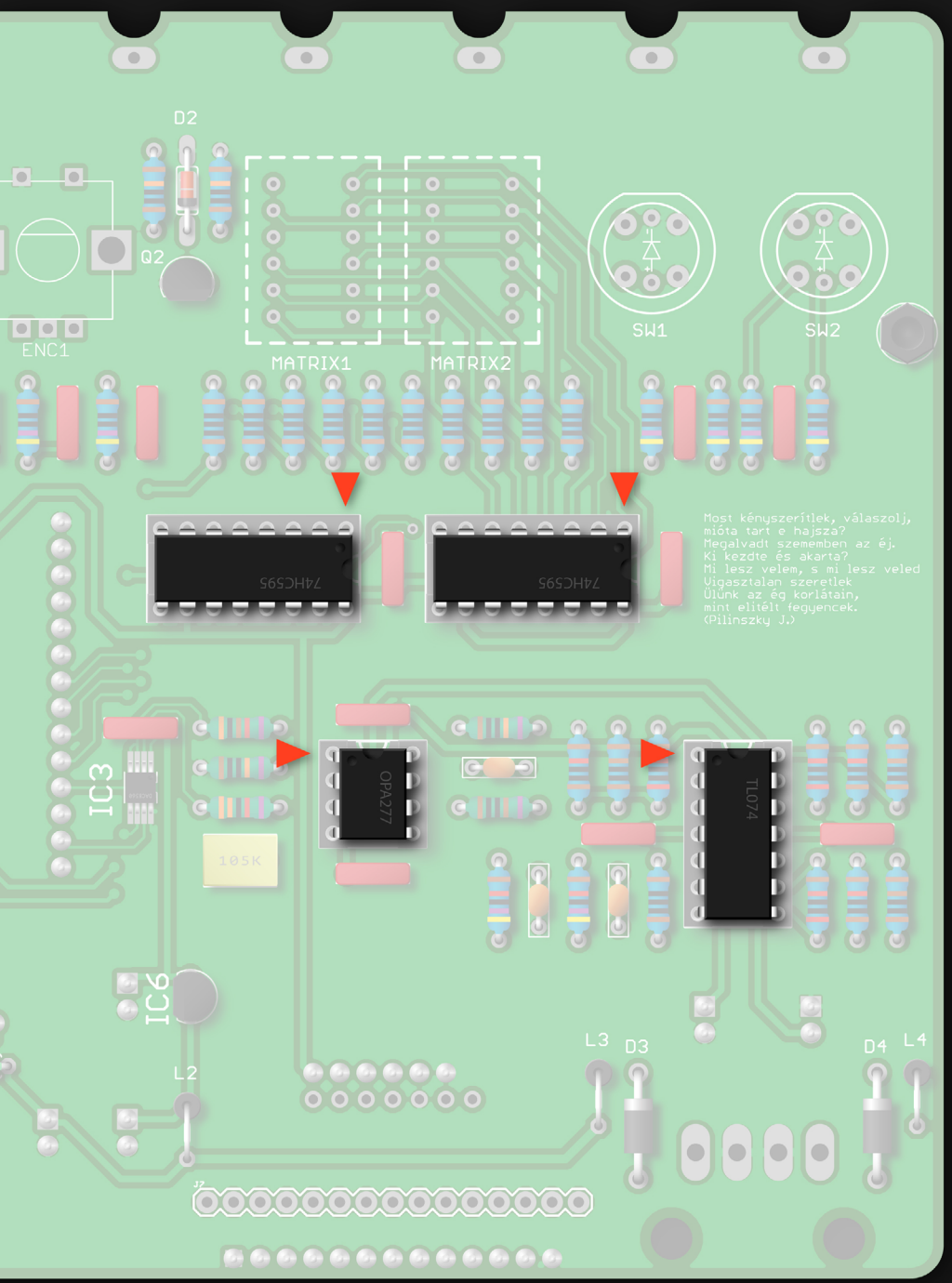
TSNM MKII DOES NOT WORK WITH TEENSY 4.0!

#29 / MISC. COMPONENTS

2x 10mm SPACER

SPACERS SHOULD HAVE INTERNAL THREAD ON ONE END AND EXTERNAL ON THE OTHER.
MOUNT THEM TO THE PCB WITH M3 NUTS, LATER TO THE FRONTPANEL WITH M3 SCREWS.





#30 / MISC. COMPONENTS

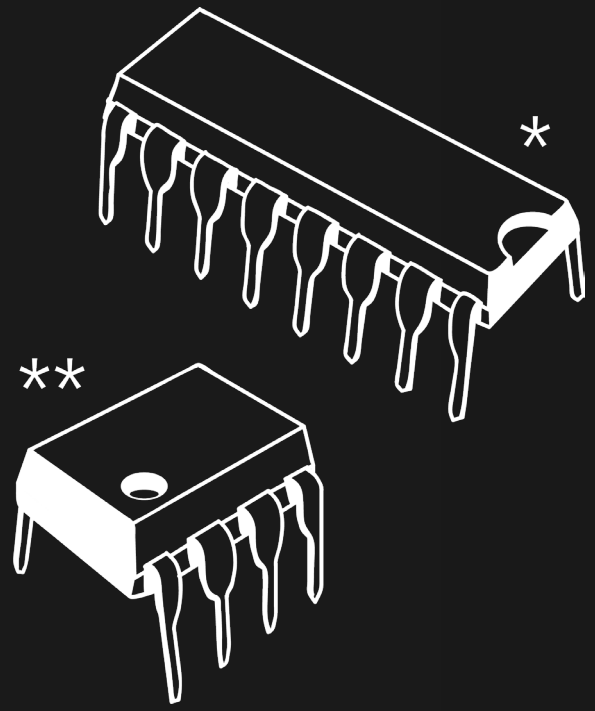
2x 74HC595 IC1, IC2

ORIENTATION: THE NOTCH* SHOULD BE ON THE RIGHT
OR THE CIRCULAR RECESS** IN THE TOP RIGHT CORNER (AS PICTURED)

1x OPA272 IC3

1x TL074 IC4

ORIENTATION: THE NOTCH* SHOULD BE ON THE TOP
OR THE CIRCULAR RECESS** IN THE TOP LEFT CORNER (AS PICTURED)

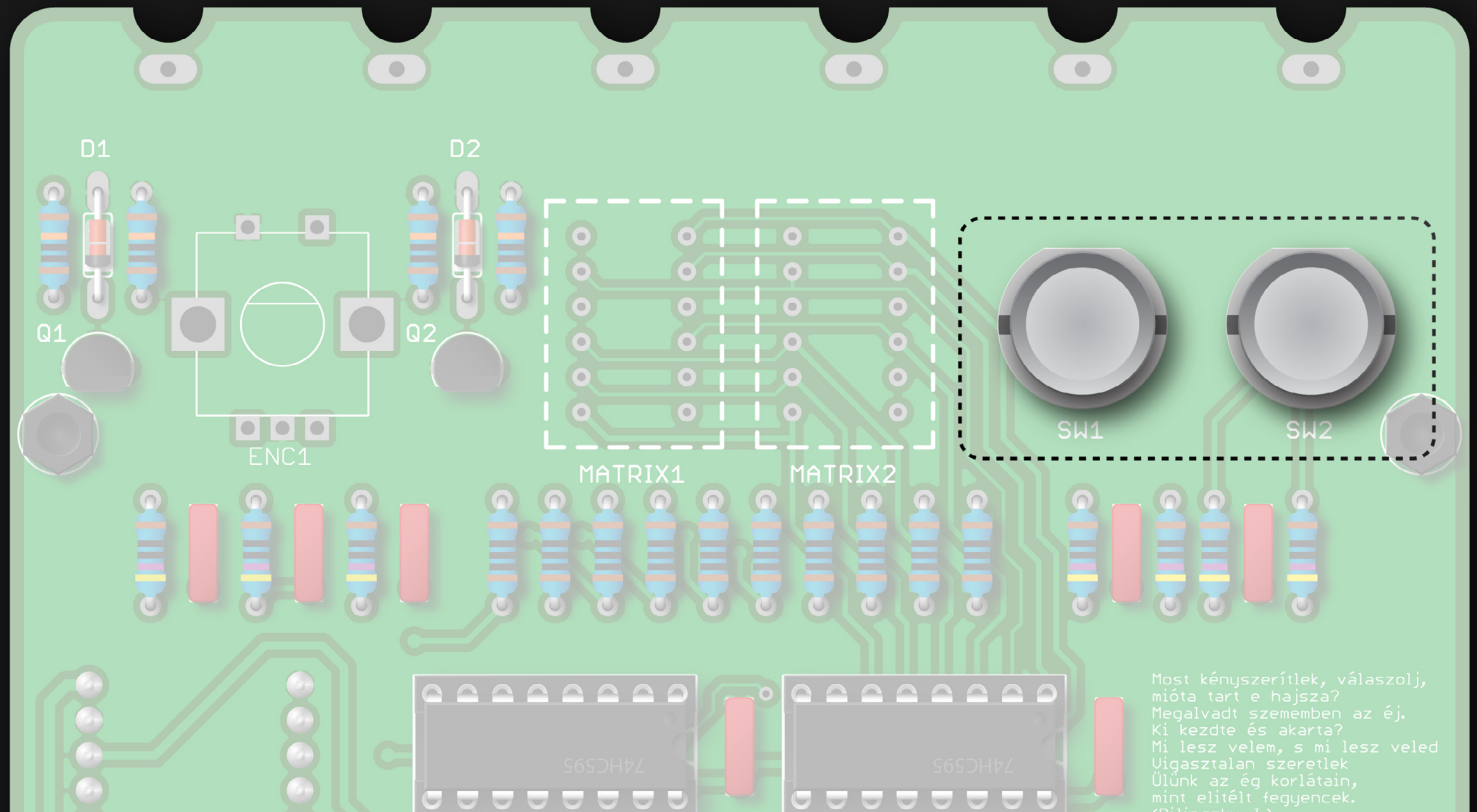


#31 / MISC. COMPONENTS

2x ILLUMINATED PUSH-BUTTON

SW1, SW2

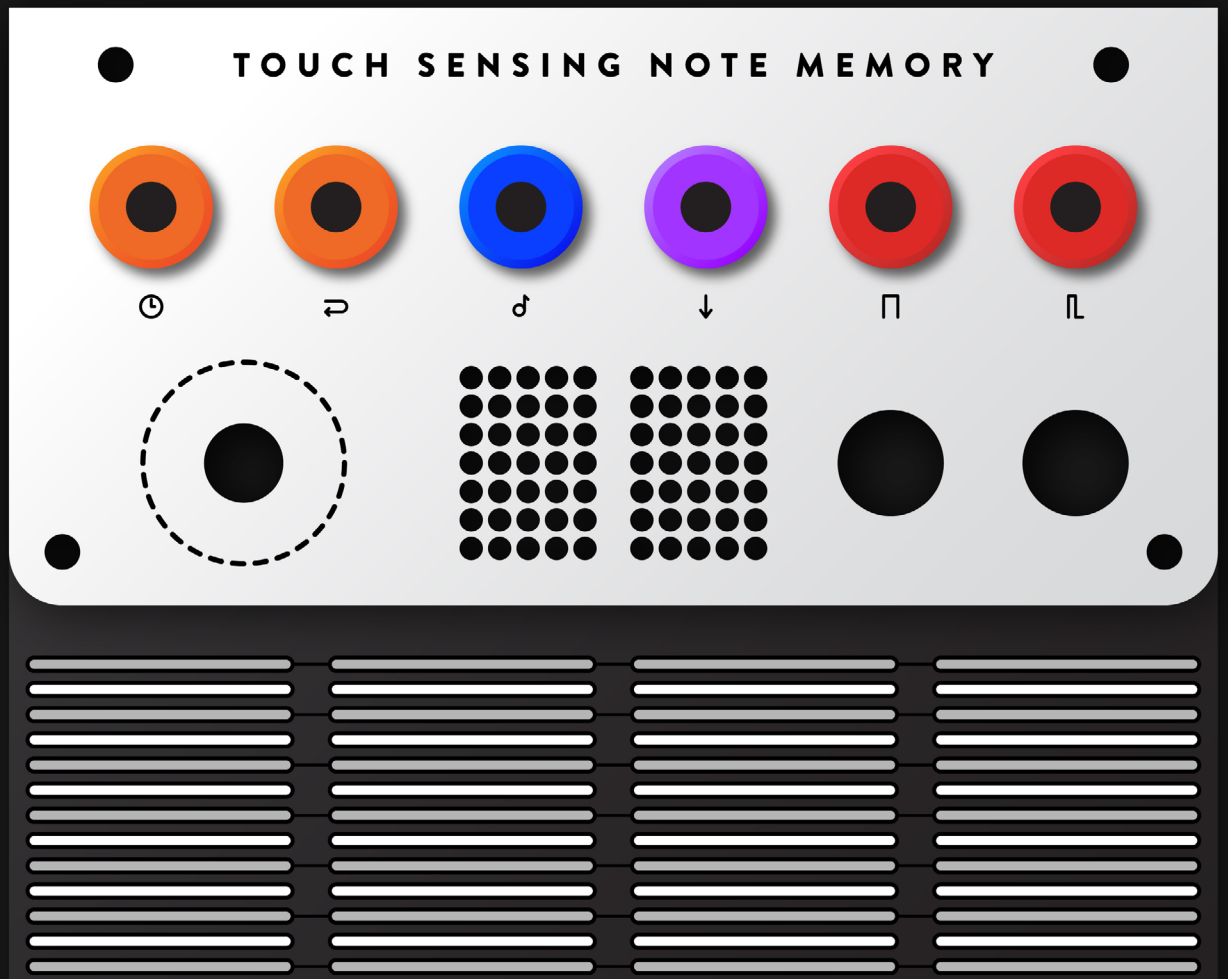
ORIENTATION: **MATCH UP WITH THE SILKSCREEN MARKING, THE FLATTENED SIDE SHOULD BE ON THE TOP!**
APPLY GENTLE PRESSURE WHILE SOLDERING TO MAKE SURE THAT THE PUSH-BUTTON SITS FLUSH ON THE PCB.



#32 / MISC. COMPONENTS

PREPARE THE FRONTPANEL

PLACE THE ANODISED ALUMINIUM PANEL ON THE MAIN FRONTPANEL, PUT IN PLACE THE BANANA CONNECTORS, RE-CHECK THE ALIGNMENT OF THE PANELS, TIGHTEN THE NUTS.



#33 / MISC. COMPONENTS

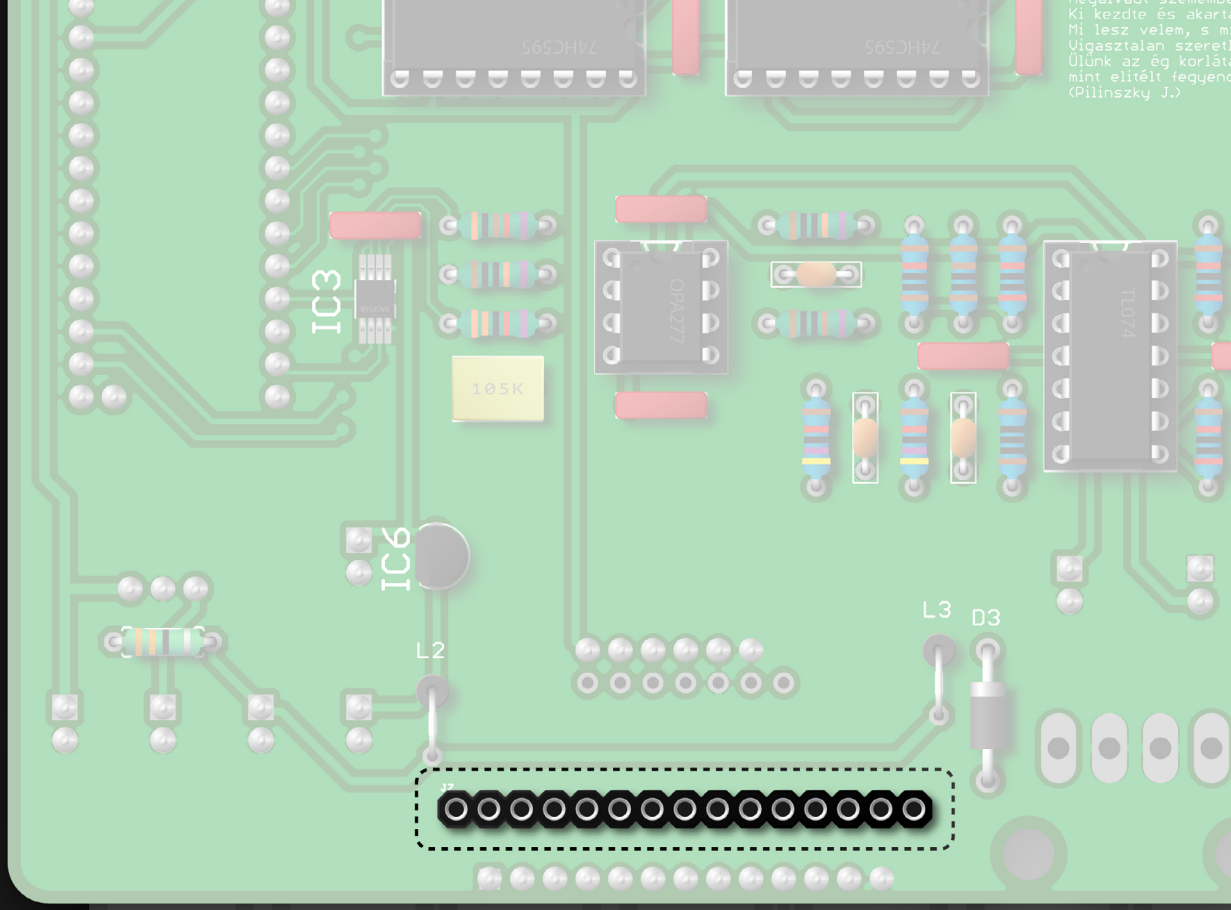
1x 15-PIN FEMALE HEADER

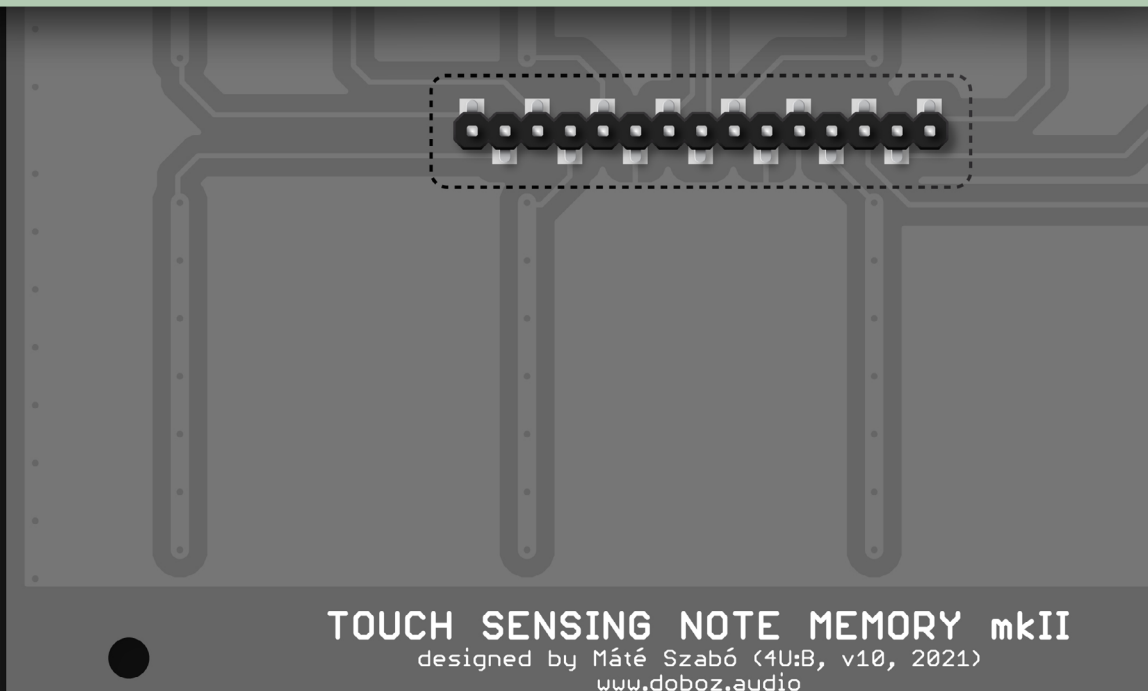
(J7 ON THE TOP SIDE OF PCB)

1x 15-PIN SMD MALE HEADER

(ON THE BACK OF THE FRONTPANEL)

- SLIDE THE FRONT PANEL ONTO THE BUSHBUTTONS, TIGHTEN THE SCREWS INTO THE SPACERS
- MAKE SURE THE SMD HEADER IS STANDING IN IT'S PLACE ON THE BACKSIDE OF THE FRONT PANEL
- START BY SOLDERING THE FEMALE HEADER TO THE PCB
- SOLDER THOSE PINS OF THE SMT HEADER THAT ARE VISIBLE
- REMOVE THE FRONT PANEL, PAY ATTENTION TO THE 15 PIN SMD MALE HEADER, PULL IT APART GENTLY, NOT TO RIP OFF THE PADS AND TRACES FROM THE FRONT PANEL
- SOLDER THE REMAINING PINS OF THE SMD HEADER





TOUCH SENSING NOTE MEMORY mkII

designed by Máté Szabó (4U:B, v10, 2021)

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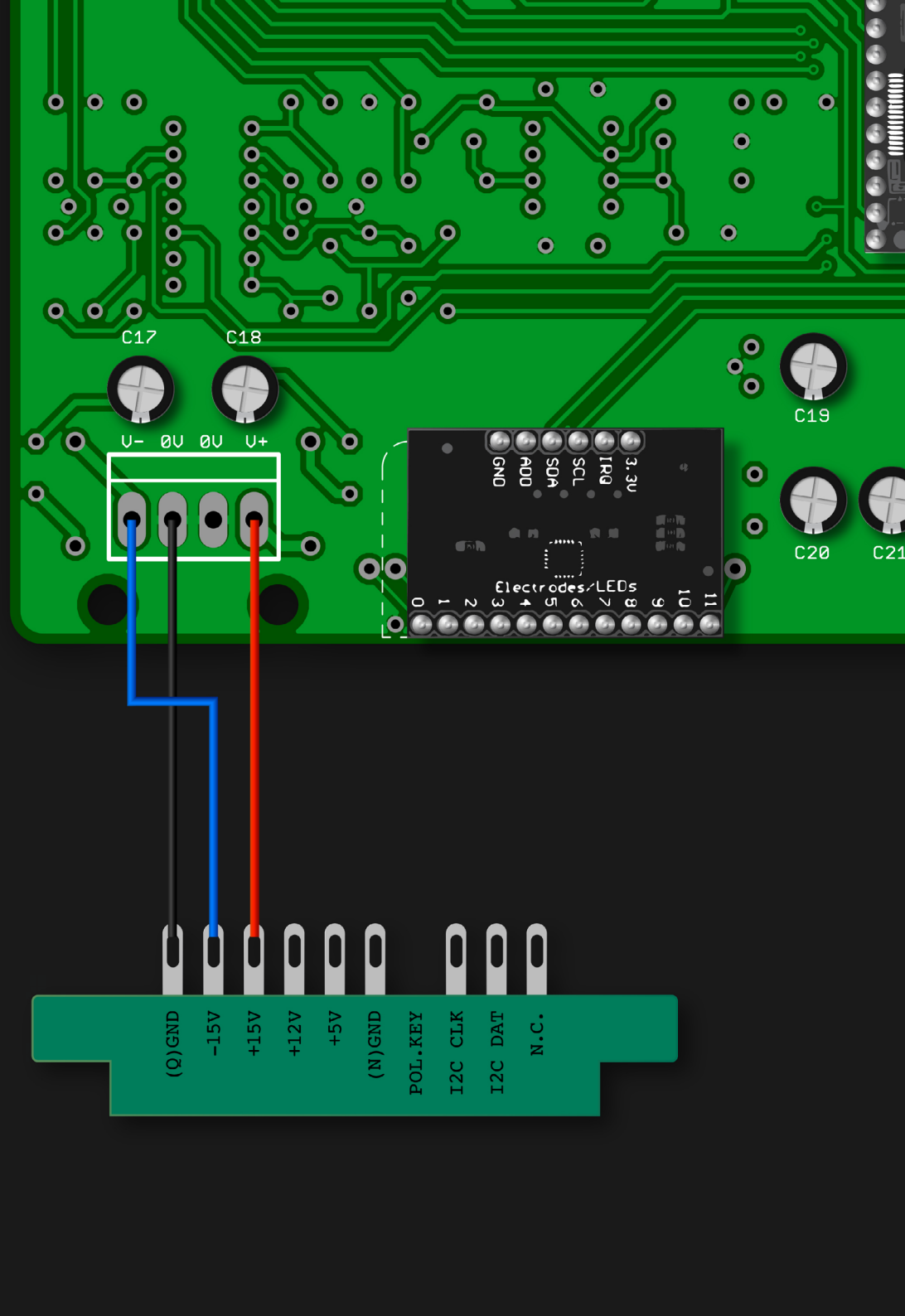
#34 / MISC. COMPONENTS

POWER CONNECTOR

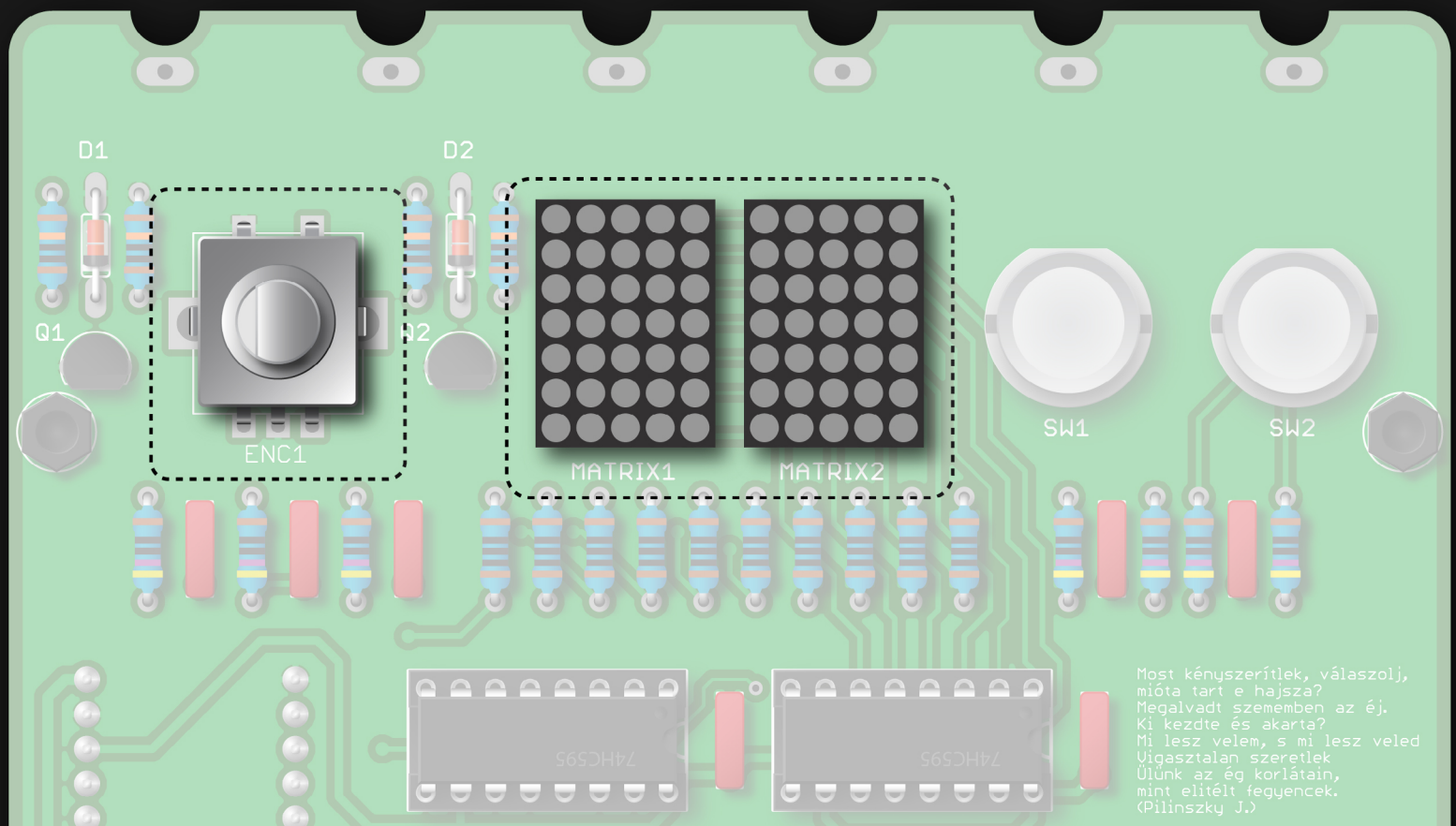
WIRE UP THE EDAC CONNECTOR TO THE PCB ACCORDING TO THE ILLUSTRATION. USE CCA. 35 CM LONG, 22AWG STRANDED WIRES. INSTALL A POLARIZING KEY (OR SOLDER A BLOB OF SOLDER) TO PIN 7. FASTEN THE CABLES TO THE PCB WITH A CABLE TIE.



2x LED DISPLAY MATRIX1, MATRIX2
1x ENCODER ENC1

- INSERT THE MATRICES AND THE ENCODER, DON'T SOLDER THEM YET
- TO LINE UP THE ENCODER WITH THE SPACERS MOUNT AN EXTRA NUT OR WASHERS ON ITS THREAD (THIS WILL NARROW THE GAP BETWEEN THE ENCODER AND THE PANEL)
- SLIDE THE FRONT PANEL ONTO ITS PLACE, TIGHTEN THE SCREWS INTO THE SPACERS
- TIGHTEN THE NUT ONTO THE ENCODERS THREAD
- ALIGN THE MATRICES THAT THEY LINE UP WITH THE HOLES ON THE FRONT PANEL AS ACCURATELY AS POSSIBLE
- NOW CAREFULLY SOLDER THE PREPARED COMPONENTS, RE-CHECK ALIGNMENT BEFORE MOVING TO ANOTHER PIN
- TIGHTEN THE NUTS
- PLACE THE ENCODER KNOB

ORIENTATION OF VARIOUS LED MATRICES:
- **FYM-7571AUHR-11** TEXT SHOULD FACE ENCODER
- **LTP-747KR** TEXT SHOULD FACE JACKS
- **TA07-11SURKWA** TEXT SHOULD FACE ENCODER
- **KWM-20571CSB** TEXT SHOULD FACE ENCODER



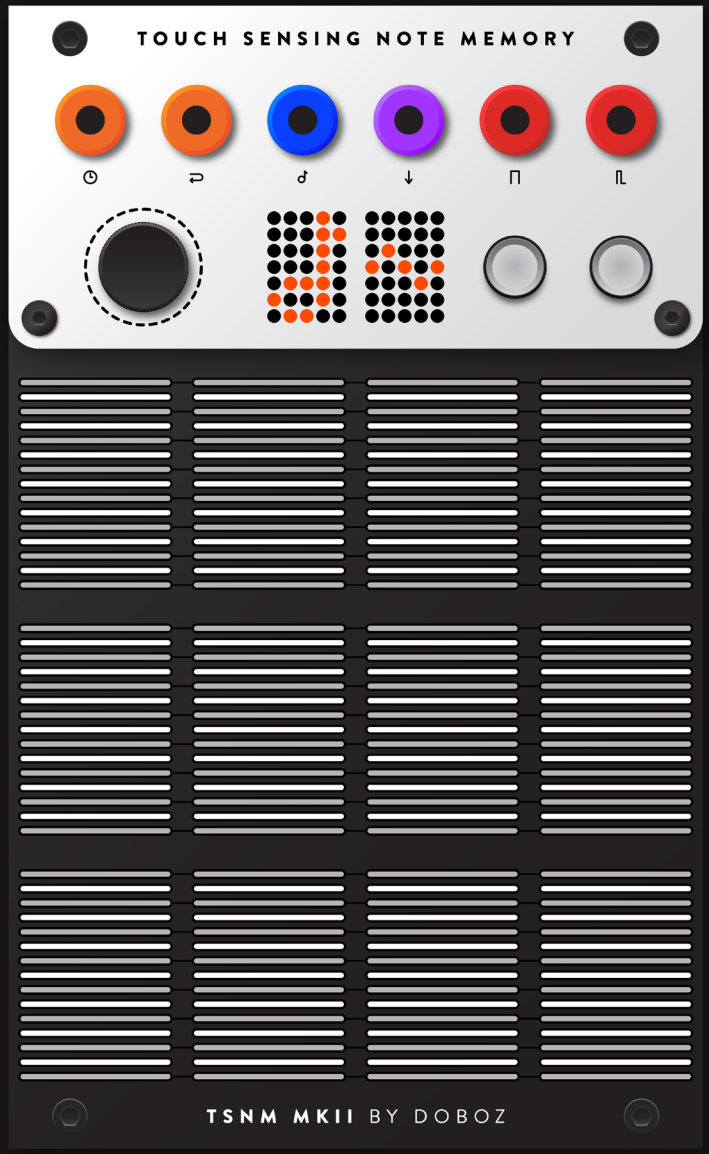
#36 / MISC. COMPONENTS

BANANA CONNECTORS

INSERT A BARE WIRE INTO THE PAD AND SOLDER THAT WIRE TO THE TOP OF THE BANANA JACK.

#37 / FIRMWARE UPLOAD

- DOWNLOAD THE ZIP FILE FROM WWW.DOBOZ.AUDIO
- UNZIP FILES, FOLLOW THE INSTRUCTIONS



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